GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

Teacher's Edition

The Period

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must Add and subtract numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not The two exponents stay within 2 of each other — that is this benchmark's own limit. Multiplication and division are 4.02–03; significant digits are 4.04.

Target I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

MTR **MTR.5.1** — Notes I · **MTR.6.1** — Boards 3 and 4 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The powers of ten have to match before a single digit is added. That is the whole lesson.
2. The power of ten is a label, like the word thousands. You never add the label.
3. When you finish, look at the coefficient. If it is not between 1 and 10, you are not done.

Slide by Slide

1. **Adding and Subtracting in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is today's lesson in miniature. *1 retrieves 3.09 — small numbers in scientific notation: the 6 is in the ten-thousandths place (yesterday). 2 retrieves 3.08 — divide the coefficients, divide the powers, multiply the results (last week). 3 retrieves Unit 2 — perfect squares (2.01). Roots come back for good in 4.05 (last unit). 4 retrieves MA.5.NSO.2.3 — adding decimals: line up the place values, not the digits (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 6.2×10^{-4} 2. 4,000 3. 12 4. 1.69
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** You can only add things that are the same size. Ten-thousands add to ten-thousands, and the label never changes.
6. **Notes.** The power of ten is a label. Six thousands plus two thousands is eight thousands — you never add the word thousands.
7. **Notes.** The exponents are never added. Adding 10 to the fifth and 10 to the third does not make 10 to the eighth — it does not make a power of ten at all.
8. **Example 1 — Same power, and the answer is not finished.** SPOKEN ONLY.
9. **Example 1 — Worked.** Add the coefficients, keep 10^7, then repair.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape with negative exponents.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Different powers, and a subtraction.** SPOKEN ONLY.
13. **Example 2 — Worked.** Rewrite 4.5×10^5 over 10^6 — exponent up one, point one place left — and the subtraction becomes ordinary.
14. **Your Turn — Same steps, your numbers. Boards up when done.** The exponents are two apart — the widest gap this benchmark allows.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. $(3.2 \times 10^5) + (4.1 \times 10^5)$ → ***7.3×10^5*** *7.3×10^{10} — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]; 1.312×10^{11} — the numbers were multiplied [8.NSO.1.5: the operation is addition]*

18–19. Board 2. A stadium sold 4.2×10^4 tickets on Friday and 3.5×10^4 on Saturday. Each ticket cost 38 dollars. How many tickets were sold on the two days? → ***7.7×10^4*** *The ticket price of 38 dollars is not needed. anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]; 7.7×10^8 — the exponents were added [B1G-M 8.NSO.1.5 misconception]*

20–21. Board 3. $(8 \times 10^{-6}) + (3 \times 10^{-6})$ → ***1.1×10^{-5}*** *11×10^{-6} — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; 1.1×10^{-11} — the exponents were added, giving 11×10^{-12}, and the answer was then finished*

22–23. Board 4. $(5.6 \times 10^8) - (4.9 \times 10^8)$ → ***7×10^7*** *0.7×10^8 — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; 1.05×10^9 — added instead of subtracting*

24–25. Board 5. $(1.6 \times 10^4) + (2.3 \times 10^6)$. Which is the sum? → ***C — 2.316×10^6*** *(A) the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]; (B) the coefficients were added and the exponents were added [B1G-M 8.NSO.1.5 named misconception: $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$]; (D) 1.6×10^4 was rewritten as 0.16×10^6 — the decimal point moved one place too few [8.NSO.1.5: the point moves as many places as the exponent changes] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. $(2.5 \times 10^6) + (9.8 \times 10^6)$. Which is the sum, written in scientific notation? → ***D — 1.23×10^7*** *(A) equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (B) the exponents were added as well as the coefficients [8.NSO.1.5: the power of ten does not change]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$ → ***7.16×10^{-2}*** *1.4×10^{-2} — the coefficients were subtracted without matching the powers [8.NSO.1.5]; 7.16×10^{-5} — the exponents were added, $-2 + (-3)$ [8.NSO.1.5: the power of ten is a label, not a term]*

30–31. Board 8. $(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$. What is the value of n? → ***$n = 5.7$*** *-3.48 — the target was not rewritten over 10^6 before subtracting [8.NSO.1.5: match the powers first]; 5.7×10^6 — the blank holds a coefficient, not a whole term*

32–33. Board 9. Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$. Name the two mistakes he made, and give the correct sum. → ***6.619×10^4*** *'he added wrong' — no mistake is actually named [the item asks which two]; 8.5×10^4 — the exponent mistake was caught and the matching mistake was not*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–6 are one move each and walk the whole decision tree: same power with no repair, same power subtracting, a repair upward, a repair downward, a gap of 1, a gap of 2 with negative exponents. 7 and 8 are two real quantities each, one a sum and one a difference, so the same pair of numbers is used twice in different directions. 9 works backwards in three shapes — a missing addend, a missing subtrahend, and a sum that lands exactly on 10. The pattern breaks at 10, where the student reads someone else's work and has to name two mistakes rather than find one answer. 11 is the diagnostic multiple choice and 12 is the select-all, whose two wrong options are the two ways of misreading an exponent. The Additional sheet mirrors each position.

Misconceptions to Watch

Board 1. 7.3×10^{10} — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]

Board 2. anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]

Board 3. 11×10^{-6} — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]

Board 4. 0.7×10^8 — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]

Board 5. the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]

Board 6. equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^{9} instead of 1.2×10^{10}]

Board 7. 1.4×10^{-2} — the coefficients were subtracted without matching the powers [8.NSO.1.5]

Board 8. −3.48 — the target was not rewritten over 10^{6} before subtracting [8.NSO.1.5: match the powers first]

Board 9. 'he added wrong' — no mistake is actually named [the item asks which two]

Materials Whiteboards and markers. Calculators are allowed on everything; entering 8E−6 + 3E−6 and reading 1.1E−5 back off the display is worth thirty seconds at whiteboard 3.

Adding and Subtracting in Scientific Notation

Lesson Plan

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1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 1 — Adding and Subtracting in Scientific Notation	Date	________________
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.5

2. Standards

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must. Add and subtract numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not. The two exponents stay within 2 of each other — that is this benchmark's own limit. Multiplication and division are 4.02–03; significant digits are 4.04.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the powers of ten are set against the place-value names, so 'the powers must match' is read off the structure rather than announced as a rule.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 3 and 4 — the coefficient is checked against $1 \le a < 10$ before the board goes up, in both directions.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 1.23×10^7 and 12.3×10^6, which have the same value, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

Essential question. Why must two numbers written in scientific notation carry the same power of ten before they can be added?

Building on. MA.8.NSO.1.4 (3.08–3.09) — scientific notation, the coefficient rule and reading notation back; Grade 6 — adding and subtracting decimals by place value.

Working toward. 4.02–03 — multiplying and dividing in scientific notation; MA.8.NSO.1.6 (4.04) — real-world problems and significant digits.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 4 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why can 5.2×10^6 and 3.1×10^6 be added straight away, while 5.2×10^6 and 3.1×10^5 cannot?	DOK 2
Devon adds the exponents and gets 10^6. What would have to be true about the expression for adding exponents to be the right move?	DOK 3
12.3×10^6 and 1.23×10^7 are the same number. Why does one of them count as an answer and the other does not?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$(3.2 \times 10^5) + (4.1 \times 10^5)$	***7.3×10^5***	7.3×10^{10} — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]; 1.312×10^{11} — the numbers were multiplied [8.NSO.1.5: the operation is addition]
2	A stadium sold 4.2×10^4 tickets on Friday and 3.5×10^4 on Saturday. Each ticket cost 38 dollars. How many tickets were sold on the two days?	***7.7×10^4***	anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]; 7.7×10^8 — the exponents were added [B1G-M 8.NSO.1.5 misconception]
3	$(8 \times 10^{-6}) + (3 \times 10^{-6})$	***1.1×10^{-5}***	11×10^{-6} — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; 1.1×10^{-11} — the exponents were added, giving 11×10^{-12}, and the answer was then finished
4	$(5.6 \times 10^8) - (4.9 \times 10^8)$	***7×10^7***	0.7×10^8 — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; 1.05×10^9 — added instead of subtracting
5	$(1.6 \times 10^4) + (2.3 \times 10^6)$. Which is the sum?	***C — 2.316×10^6***	(A) the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]; (B) the coefficients were added and the exponents were added [B1G-M 8.NSO.1.5 named misconception: $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$]; (D) 1.6×10^4 was rewritten as 0.16×10^6 — the decimal point moved one place too few [8.NSO.1.5: the point moves as many places as the exponent changes]

#	What the board asks	Answer	If the board is wrong
6	$(2.5 \times 10^6) + (9.8 \times 10^6)$. Which is the sum, written in scientific notation?	***D — 1.23×10^7***	(A) equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (B) the exponents were added as well as the coefficients [8.NSO.1.5: the power of ten does not change]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]
7	$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$	***7.16×10^{-2}***	1.4×10^{-2} — the coefficients were subtracted without matching the powers [8.NSO.1.5]; 7.16×10^{-5} — the exponents were added, $-2 + (-3)$ [8.NSO.1.5: the power of ten is a label, not a term]
8	$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$. What is the value of n?	***$n = 5.7$***	-3.48 — the target was not rewritten over 10^6 before subtracting [8.NSO.1.5: match the powers first]; 5.7×10^6 — the blank holds a coefficient, not a whole term
9	Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$. Name the two mistakes he made, and give the correct sum.	***6.619×10^4***	'he added wrong' — no mistake is actually named [the item asks which two]; 8.5×10^4 — the exponent mistake was caught and the matching mistake was not

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the place-value strip from Notes I with the power written above each name, and let the student write both numbers as words first — 'four point nine five ten-thousands' and 'twelve point one ten-thousands' — before any symbol is written. The matching step becomes reading, not procedure.

ELL. Coefficient and power of ten are the two phrases every question uses. Write 7.7×10^4 once with COEFFICIENT under the 7.7 and POWER OF TEN under the 10^4, and point at the labels on every board. Add or subtract is the third phrase: read each board's operation aloud before the boards go up.

Enrichment. Ask for $(4.2 \times 10^6) + (3.9 \times 10^5) + (1.1 \times 10^4)$ — three terms, all within two of each other — and then why the benchmark stops at a gap of two.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name both of Devon's mistakes separately and give 6.619×10^4. Read the boards, not the papers.

Vocabulary. **scientific notation**; **coefficient**; **common power of ten**

Materials. Whiteboards and markers. Calculators are allowed on everything; entering 8E−6 + 3E−6 and reading 1.1E−5 back off the display is worth thirty seconds at whiteboard 3.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 4 test: Add and subtract numbers written in scientific notation — HUR.

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

Teacher's Edition

The Period

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

Target I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

MTR **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 3 and 7 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	12 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The powers of ten never have to match when you multiply or divide. They can be as far apart as they like.
2. Multiply the fronts and add the backs; divide the fronts and subtract the backs, top minus bottom.
3. A product of two coefficients under 10 is very often over 10. Look at the coefficient before you put the board up.

Slide by Slide

1. **Multiplying and Dividing in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q1 and Q2 are the two halves of today, side by side. *1 retrieves 4.01 — same power: add the coefficients and keep the power (yesterday). 2 retrieves 3.02 — quotient of powers with base 10: subtract the exponents (last week). 3 retrieves Unit 2 — perfect squares (2.01). Roots return in 4.05 (last unit). 4 retrieves MA.6.NSO.2.1 — dividing decimals; today every quotient starts with one of these (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 6.2×10^6 2. 10^5 3. 13 4. 3
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy both lines.** Multiply the fronts, add the backs. The powers of ten never have to match when you multiply.
6. **Notes.** Top minus bottom, always in that order. And subtracting a negative goes up.
7. **Notes.** Coefficient over 10, point left and exponent up. Coefficient under 1, point right and exponent down. Same repair as yesterday.
8. **Example 1 — Multiplying, and the answer needs finishing.** SPOKEN ONLY.
9. **Example 1 — Worked.** Regroup, multiply, add the exponents, then repair.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the other sign on the small exponent.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Multiplication and division in one expression.** SPOKEN ONLY.
13. **Example 2 — Worked.** The intermediate value is 60×10^6, which is exactly the unfinished form — and it does not have to be repaired yet, because the division is still to come.
14. **Your Turn — Same steps, your numbers. Boards up when done.** A negative exponent inside the product and a division that pushes the answer below 1.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. $(3\times10^5)(3\times10^6)$ → ***9×10^{11}*** *9×10^{30} — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; 6×10^{11} — the coefficients were added [8.NSO.1.5: the operation is multiplication]*

18–19. Board 2. A printer makes 4×10^3 copies per hour. It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet. How many copies does it make? → ***1×10^6*** *The paper cost of 3 cents a sheet is not needed. anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; 10×10^5 — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]*

20–21. Board 3. $\frac{9.6\times10^8}{1.2\times10^3}$ → ***8×10^5*** *8×10^{11} — the exponents were added [8.NSO.1.5: division subtracts the exponents]; 8.4×10^5 — the coefficients were subtracted instead of divided [8.NSO.1.5]*

22–23. Board 4. $\frac{3.28\times 10^{7}}{4\times 10^{10}}$ → ***8.2×10^{-4}*** *0.82×10^{-3} — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; 8.2×10^{-3} — the point moved but the exponent did not*

24–25. Board 5. $(4 \times 10^{7})(2 \times 10^{9})$. Which is the product? → ***C — 8×10^{16}*** *(A) the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^{4})(3 \times 10^{5}) = 6 \times 10^{20}$]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. $(5 \times 10^{6})(4 \times 10^{3})$. Which is the product, written in scientific notation? → ***D — 2×10^{10}*** *(A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^{9} instead of 1.2×10^{10}]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $\frac{7\times 10^{9}}{7\times 10^{-6}}$ → ***1×10^{15}*** *1×10^{3} — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; 1×10^{-15} — bottom minus top [8.NSO.1.5: top minus bottom]*

30–31. Board 8. $(8.4 \times 10^{7}) \div (n \times 10^{2}) = 4.2 \times 10^{5}$. What is the value of n? → ***n = 2*** *4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]; 0.5 — 4.2 ÷ 8.4 instead of 8.4 ÷ 4.2*

32–33. Board 9. For $(7.2 \times 10^{11}) \div (2.4 \times 10^{3})$, Marcus writes 4.8×10^{8} and Sara writes 3×10^{14}. Name what each one did wrong, and give the correct quotient. → ***3×10^{8}*** *'they both got it wrong' — neither error is named [the item asks what each one did]; 4.8×10^{8} — Marcus's error repeated*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–6 are one move each and alternate multiply, divide, multiply, divide: a plain product, a plain quotient, a product that must be finished upward, a quotient that must be finished downward, a product of two negative exponents, a quotient of two negative exponents. 7 and 8 are real quantities — 7 is the state guide's meteorite set, used only for ratios, and 8 is a rate in both directions, total from a rate and rate from a total. 9 works backwards in three shapes: a missing exponent, a missing coefficient, and a product landing exactly on 10. The pattern breaks at 10, where the student judges someone else's correct value in an unfinished form. 11 is the diagnostic multiple choice and 12 the select-all, whose two wrong options are the two exponent errors. The Additional sheet mirrors each position.

Misconceptions to Watch

Board 1. 9×10^{30} — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^{4})(3 \times 10^{5}) = 6 \times 10^{20}$]

Board 2. anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]

Board 3. 8×10^{11} — the exponents were added [8.NSO.1.5: division subtracts the exponents]

Board 4. 0.82×10^{-3} — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \leq a < 10$]

Board 5. the exponents were subtracted, $7 - 9$ [8.NSO.1.5: multiplication adds the exponents]

Board 6. the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]

Board 7. 1×10^{3} — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]

Board 8. 4.2 — the coefficients were subtracted, $8.4 - 4.2$ [8.NSO.1.5: the coefficients divide]

Board 9. 'they both got it wrong' — neither error is named [the item asks what each one did]

Materials Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether 8×10^{16} and 8×10^{63} could possibly both be the answer to whiteboard 5.

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

Lesson Plan

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1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lessons 2–3 — Multiplying and Dividing in Scientific Notation	Date	________________
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.5

2. Standards

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must. Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not. No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the product is regrouped before it is evaluated, so 'multiply the fronts, add the backs' is read off the structure rather than memorised as a rule.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1, 3 and 7 — one move each, written and up on the cue; dividing in scientific notation was the lowest single figure on the last state report.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 2×10^{10} and 20×10^{9}, which are the same number, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

Essential question. Why must the powers of ten match to add, but not to multiply?

Building on. 4.01 — the coefficient rule and the finishing step; MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.03) — the product and quotient of powers laws.

Working toward. 4.04 — real-world problems and significant digits; MA.912.NSO.1 — operations with rational exponents in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 4 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why do the powers of ten have to match before you add, but not before you multiply?	DOK 3
Marcus's power of ten is right and Sara's coefficient is right. Which of the two is closer to the correct answer, and does 'closer' mean anything useful here?	DOK 3
A quotient of two positive numbers comes out as 5.3×10^{-3}. What does the negative exponent tell you about how the two numbers compared?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$(3\times10^5)(3\times10^6)$	***9×10^{11}***	9×10^{30} — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; 6×10^{11} — the coefficients were added [8.NSO.1.5: the operation is multiplication]
2	A printer makes 4×10^3 copies per hour. It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet. How many copies does it make?	***1×10^6***	anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; 10×10^5 — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]
3	$\dfrac{9.6\times10^8}{1.2\times10^3}$	***8×10^5***	8×10^{11} — the exponents were added [8.NSO.1.5: division subtracts the exponents]; 8.4×10^5 — the coefficients were subtracted instead of divided [8.NSO.1.5]
4	$\dfrac{3.28\times10^7}{4\times10^{10}}$	***8.2×10^{-4}***	0.82×10^{-3} — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; 8.2×10^{-3} — the point moved but the exponent did not
5	$(4 \times 10^7)(2 \times 10^9)$. Which is the product?	***C — 8×10^{16}***	(A) the exponents were subtracted, $7 - 9$ [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication]
6	$(5 \times 10^6)(4 \times 10^3)$. Which is the product, written in scientific notation?	***D — 2×10^{10}***	(A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5]
7	$\dfrac{7\times10^9}{7\times10^{-6}}$	***1×10^{15}***	1×10^3 — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; 1×10^{-15} — bottom minus top [8.NSO.1.5: top minus bottom]
8	$(8.4 \times 10^7) \div (n \times 10^2) = 4.2 \times 10^5$. What is the value of n?	***$n = 2$***	4.2 — the coefficients were subtracted, $8.4 - 4.2$ [8.NSO.1.5: the coefficients divide]; 0.5 — $4.2 \div 8.4$ instead of $8.4 \div 4.2$
9	For $(7.2 \times 10^{11}) \div (2.4 \times 10^3)$, Marcus writes 4.8×10^8 and Sara writes 3×10^{14}. Name what each one did wrong, and give the correct quotient.	***3×10^8***	'they both got it wrong' — neither error is named [the item asks what each one did]; 4.8×10^8 — Marcus's error repeated

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give a two-column organiser — COEFFICIENTS on the left, POWERS OF TEN on the right — and have the student split every expression into the two columns before computing anything. The regrouping step in Notes I becomes a physical act, and the two laws are then applied one column at a time.

ELL. Product and quotient are the two nouns, multiply and divide the two verbs. Write them in pairs on the board — PRODUCT goes with MULTIPLY and ADD the exponents, QUOTIENT goes with DIVIDE and SUBTRACT the exponents — and read each board's operation aloud before the boards go up.

Enrichment. Ask for $(4 \times 10^5)^2$ without a calculator (1.6×10^{11}), then for the two numbers in scientific notation whose product is 1.2×10^7 and whose quotient is 1.2×10^1 (1.2×10^4 and 1×10^3).

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name Marcus's error and Sara's error separately and give 3×10^8. Read the boards, not the papers.

Vocabulary. **product of powers law**; **quotient of powers law**; **regrouping a product**

Materials. Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether 8×10^{16} and 8×10^{63} could possibly both be the answer to whiteboard 5.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 4 test: Multiply numbers written in scientific notation — YZU; Divide numbers written in scientific notation — SGT.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Teacher's Edition

The Period

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Must Solve real-world problems with numbers in scientific notation, and recognise the importance of significant digits when physical measurements are involved.

Must not The within-2 exponent limit still applies to addition and subtraction here. Significant digits limit a reported answer; they never change the arithmetic, and a count is never a measurement.

Target I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

MTR **MTR.6.1** — Boards 2, 6 and 7 · **MTR.7.1** — Every problem from Notes II on · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A measurement's last digit is uncertain. That is what the whole rule is protecting.
2. Products and quotients count digits. Sums and differences count places. Two rules, and it matters which one you reach for.
3. Do all the arithmetic first and round once, at the end. Rounding in the middle loses digits the arithmetic still needed.

Slide by Slide

1. **Real-World Problems and Significant Digits.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q3 and Q4 are today's two tools. *1 retrieves 4.02–03 — multiply the coefficients, add the exponents, then finish the answer (yesterday). 2 retrieves 4.01 — rewrite one number over the other's power, then subtract the coefficients (last week). 3 retrieves 3.09 — small numbers; note that the zero between the 9 and the 2 stays (last unit). 4 retrieves MA.5.NSO.1.5 — rounding to a named place; every answer today ends with one of these (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 1.2×10^4 2. 5.89×10^7 3. 9.02×10^{-3} 4. 4.74
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** Scientific notation makes this easy. In 5.2 times ten to the fourth, every digit you can see is significant — that is one of the reasons the notation exists.
6. **Notes.** Two significant digits in, two significant digits out. The answer is never better than the measurement it came from.
7. **Notes — Copy all three lines.** Products count digits. Sums count places. They are two different rules and it matters which one you reach for.
8. **Example 1 — A product of two measurements.** SPOKEN ONLY.
9. **Example 1 — Worked.** Multiply in full — 10,750 — and round once.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, the counts the other way round.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A difference of two measurements.** SPOKEN ONLY.
13. **Example 2 — Worked.** Subtract in full, then stop at the leftmost last-digit place.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 3,850 stops at the tens place and 274 stops at the ones, so the answer stops at the tens.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. How many significant digits are in 0.00470 ? → ***3*** *2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; 5 — the leading zeros were counted [B1G-M 8.NSO.1.6: leading zeros are never significant]*

18–19. Board 2. A pond's surface measures 1.4×10^2 m by 8.5×10^1 m. The pond is 3 meters deep. Find the surface area. Round to the correct number of significant digits. → ***1.2×10^4 square meters*** *The depth of 3 meters is not needed. anything built from the depth — the unneeded figure was used [an area needs two dimensions]; 1.19×10^4 — the arithmetic without the rounding [8.NSO.1.6 clarification]*

20–21. Board 3. Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5. How should the answer be written? → ***7.41×10^5*** *7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]; 7.4126×10^5 — no rounding at all*

22–23. Board 4. $(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured. Write the sum with the correct precision. → ***3.0×10^3 m*** *2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]; 2.9×10^3 — 2,971 rounded down instead of up [rounding to the hundreds place]*

24–25. Board 5. How many significant digits are in 0.03080? → ***A — 4*** *(B) only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]; (C) one of the leading zeros was counted [B1G-M 8.NSO.1.6: leading zeros are never significant]; (D) the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm. Which is the area, to the correct significant digits? → ***B — 2.6×10^5 cm²*** *(A) the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (C) the exponent was not raised when the coefficient was repaired [8.NSO.1.5: point one place left, exponent up one]; (D) the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A car travels 4.20×10^2 km in 3.0 hours. What is its average speed, to the right number of significant digits? → ***1.4×10^2 km/h*** *140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]; 1.4×10^1 — the exponent is one too low*

30–31. Board 8. A measured length times a measured width gives an area of 3.6×10^3 m². The length measured 9.0×10^1 m. What did the width measure, to two significant digits? → ***4.0×10^1 m*** *4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]; 4.0×10^2 — the exponent is one too high*

32–33. Board 9. Ellis says that because 9,145 has 4 significant digits and 4.8 has 2, the sum 9,145 + 4.8 must be written with 2 significant digits, as 9.1×10^3. Explain why he is wrong, and give the correct sum. → ***9,150 = 9.150×10^3*** *'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]; 9,149.8 — no rounding at all, which claims a tenth that 9,145 never measured*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–4 are the four counting rules, one each: a middle zero, leading zeros, trailing zeros with a decimal point, trailing zeros without one. 5–8 are the four operations in order — a product, a quotient, a sum, a difference — so the two rules alternate; 7 is the one that shows a sum keeping more significant digits than either measurement — three, from two two-digit measurements. 9 and 10 are real quantities, and 10 is built on the count-versus-measurement distinction. The pattern breaks at 11, where the arithmetic is right and the answer is wrong. 12 is the diagnostic multiple choice, stacking the coefficient repair on the rounding. The Additional sheet mirrors each position, and its question 11 is the case where the two rules agree — which is why the wrong one survives.

Misconceptions to Watch

Board 1. 2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]

Board 2. anything built from the depth — the unneeded figure was used [an area needs two dimensions]

Board 3. 7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]

Board 4. 2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]

Board 5. only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]

Board 6. the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]

Board 7. 140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]

Board 8. 4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]

Board 9. 'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]

Materials Whiteboards and markers. Calculators are allowed on everything and are genuinely useful today, because the point of the lesson is what to do with a long display rather than how to produce one.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Lesson Plan

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1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 4 — Real-World Problems and Significant Digits	Date	________________
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.6

2. Standards

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Must. Solve real-world problems with numbers in scientific notation, and recognise the importance of significant digits when physical measurements are involved.

Must not. The within-2 exponent limit still applies to addition and subtraction here. Significant digits limit a reported answer; they never change the arithmetic, and a count is never a measurement.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 2, 6 and 7 — the last move on every board is asking how much of the answer the measurements actually support, before the board goes up.
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Every problem from Notes II on — each number is a measured quantity with a unit, because the rule has no meaning without one.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 2.6×10^5 and 2.61×10^5, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

Essential question. How much of an answer is worth writing down, when the numbers that produced it were measured?

Building on. 4.01–4.03 — all four operations in scientific notation; Grade 5 and 6 — place value and rounding to a named place.

Working toward. 4.05–4.08 — the order of operations with exponents and radicals; MA.912.GR.4 — measurement and precision in geometry.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 4 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why does the last digit of a measurement carry less information than the digits before it?	DOK 2
A sum of two two-digit measurements can have three significant digits. How is that possible, when a product of the same two could only have two?	DOK 3
Ellis uses the product rule on a sum and throws away two digits that were measured. What is each of the two rules actually protecting?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	How many significant digits are in 0.00470 ?	***3***	2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; 5 — the leading zeros were counted [B1G-M 8.NSO.1.6: leading zeros are never significant]
2	A pond's surface measures 1.4×10^2 m by 8.5×10^1 m. The pond is 3 meters deep. Find the surface area. Round to the correct number of significant digits.	***1.2×10^4 square meters***	anything built from the depth — the unneeded figure was used [an area needs two dimensions]; 1.19×10^4 — the arithmetic without the rounding [8.NSO.1.6 clarification]
3	Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5. How should the answer be written?	***7.41×10^5***	7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]; 7.4126×10^5 — no rounding at all
4	$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured. Write the sum with the correct precision.	***3.0×10^3 m***	2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]; 2.9×10^3 — 2,971 rounded down instead of up [rounding to the hundreds place]
5	How many significant digits are in 0.03080?	***A — 4***	(B) only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]; (C) one of the leading zeros was counted [B1G-M 8.NSO.1.6: leading zeros are never significant]; (D) the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]
6	A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm. Which is the area, to the correct significant digits?	***B — 2.6×10^5 cm^2***	(A) the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (C) the exponent was not raised when the coefficient was repaired [8.NSO.1.5: point one place left, exponent up one]; (D) the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification]
7	A car travels 4.20×10^2 km in 3.0 hours. What is its average speed, to the right number of significant digits?	***1.4×10^2 km/h***	140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]; 1.4×10^1 — the exponent is one too low
8	A measured length times a measured width gives an area of 3.6×10^3 m^2. The length measured 9.0×10^1 m. What did the width measure, to two significant digits?	***4.0×10^1 m***	4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]; 4.0×10^2 — the exponent is one too high
9	Ellis says that because 9,145 has 4 significant digits and 4.8 has 2, the sum 9,145 + 4.8 must be written with 2 significant digits, as 9.1×10^3. Explain why he is wrong, and give the correct sum.	***9,150 = 9.150×10^3***	'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]; 9,149.8 — no rounding at all, which claims a tenth that 9,145 never measured

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the four counting rules as a card with one worked number each, in the order of Notes I, and let it stay on the desk through the round. On every problem the student marks the significant digits on the two measurements before doing any arithmetic, so the count is decided before it is needed.

ELL. Significant, precise and measurement are three words doing specific work today. Put MEASUREMENT = a number someone measured beside COUNT = a number someone counted, with 4.20 meters under one and 24 students under the other, and ask which kind each number is before every problem.

Enrichment. Ask what 1,000 means when it is written with no decimal point — one significant digit or four — and how a scientist would write each of the two. Then ask why the state guide's own Puerto Rico item is genuinely ambiguous.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say what the rule for a sum actually is, not only that Ellis used the wrong one, and give 9.150×10^3. Read the boards, not the papers.

Vocabulary. **significant digits**; **leading zero**; **measurement**

Materials. Whiteboards and markers. Calculators are allowed on everything and are genuinely useful today, because the point of the lesson is what to do with a long display rather than how to produce one.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 4 test: Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Evaluate expressions containing square roots, cube roots and integer exponents, simplifying radicals by factoring within the benchmark's lists.

Must not Perfect squares up to 225 and perfect cubes from −125 to 125, or fractions whose parts are each in those lists. Estimating a root that is not perfect was Unit 2. No fractional exponents.

Target I can evaluate an expression that contains square roots, cube roots and integer exponents.

MTR **MTR.5.1** — Notes I · **MTR.6.1** — Boards 3 and 6 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. The radical bar is a pair of parentheses you cannot see. Finish what is under it, then take the root.
2. A cube root of a negative number is ordinary. A square root of a negative number is not a number you have.
3. A negative exponent is still a reciprocal. Nothing about the radical changes that.

Slide by Slide

1. **Evaluating Expressions with Radicals.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q3 and Q4 are the two halves of today. *1 retrieves 4.04 — leading zeros never count; the trailing zero does, because a decimal point is written (yesterday). 2 retrieves 4.02–03 — multiply the coefficients, add the exponents (last week). 3 retrieves Unit 2 — perfect squares (2.01): 14 · 14 = 196 (last unit). 4 retrieves MA.5.AR.2.2 — multiply before you subtract; today the radical joins that list (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 3 2. 6×10^{-3} 3. 14 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy all three lines.** Whatever sits under the bar is inside a pair of parentheses you cannot see. Finish it, then take the root.
6. **Notes.** Cube roots of negative numbers are ordinary. Square roots of negative numbers are not numbers you have — that is the difference to hold on to.
7. **Notes.** A negative exponent is still a reciprocal. The radical does not change a single thing about that.
8. **Example 1 — A power and a radical in one expression.** SPOKEN ONLY.
9. **Example 1 — Worked.** Two independent pieces.
10. **Your Turn — Same steps, your numbers. Boards up when done.** The guide's tiered version: a cube root this time, and a subtraction.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A radical inside a product.** SPOKEN ONLY.
13. **Example 2 — Worked.** Three layers, innermost first: under the bar, then the root, then the multiplication, then the subtraction.
14. **Your Turn — Same steps, your numbers. Boards up when done.** A negative cube root and a product that grows.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. $\sqrt{121}$ → ***11*** *60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]; ±11 — the radical sign names the principal root only [8.NSO.1.7]*

18–19. Board 2. A square patio has an area of 169 square feet. Fencing for it costs 12 dollars a foot. How long is one side of the patio? → ***13 feet*** *The fencing price of 12 dollars a foot is not needed. anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]; 52 — the perimeter, not a side; 84.5 — the area was halved*

20–21. Board 3. $\sqrt[3]{-125}$ → ***−5*** *5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]; 'not a real number' — the square-root rule applied to a cube root [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]*

22–23. Board 4. $\sqrt{2^3+17}$ → ***5*** *25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]; anything with two separate roots — the bar groups, so it is one radicand [8.NSO.1.7]*

24–25. Board 5. √(25 + 144). Which is the value? → ***D — 13*** *(A) the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]; (B) the sum was found and the root was never taken [8.NSO.1.7]; (C) the radicand was halved instead of rooted [Unit 2, 2.02: a root is not a half] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. The cube root of −1/64. Which is the value? → ***A — −1/4*** *(B) the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]; (C) the square root of 1/64 was taken and the sign kept [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]; (D) the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $\sqrt{\frac{4}{9}}+\sqrt[3]{\frac{8}{27}}$ → ***4/3*** *12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]; 2/3 — only one of the two radicals was evaluated*

30–31. Board 8. $\sqrt{5^3-25}\cdot 10^{-3}$ → ***1/100*** *10,000 — 10^{-3} read as 10^3 [8.NSO.1.3: a negative exponent is a reciprocal]; −1/100 — the exponent's sign made the value's sign [B1G-M 8.NSO.1.3 task: −b vs b^{-1}]*

32–33. Board 9. Simon says that the cube root of −27 is not a real number, because you cannot take the root of a negative. Explain why he is wrong, and give the value. → ***−3*** *'−3' alone — the value without the reason [the item asks why he is wrong]; '3' — the sign dropped as well as the rule misread*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–6 are one root each and walk the cases: a square root, a cube root, a negative radicand, a fraction, a square root with a power under the bar, a cube root with a power under the bar. 7 and 8 each put a power and a radical in one expression — 7 in three shapes of the student's own, 8 in the three the state guide supplies. 9 works backwards in three shapes, including one where the radicand carries a term. The pattern breaks at 10, where the student judges a method rather than a number. 11 is the diagnostic multiple choice on the grouping error and 12 is the select-all, whose two wrong options are the sign of a cube root and an unfinished radicand. The Additional sheet mirrors each position; its question 10 is the same error inside a subtraction.

Misconceptions to Watch

Board 1. 60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]

Board 2. anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]

Board 3. 5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]

Board 4. 25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]

Board 5. the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]

Board 6. the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]

Board 7. 12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]

Board 8. 10,000 — 10^{-3} read as 10^{3} [8.NSO.1.3: a negative exponent is a reciprocal]

Board 9. '−3' alone — the value without the reason [the item asks why he is wrong]

Materials Whiteboards and markers, and Unit 2's reference sheet if students still want the perfect-square list. Calculators are allowed on everything; a calculator will refuse the cube root of a negative on some models, which is worth showing once and explaining rather than trusting.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Lesson Plan

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1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 5 — Evaluating Expressions with Radicals	Date	________________
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Evaluate expressions containing square roots, cube roots and integer exponents, simplifying radicals by factoring within the benchmark's lists.

Must not. Perfect squares up to 225 and perfect cubes from −125 to 125, or fractions whose parts are each in those lists. Estimating a root that is not perfect was Unit 2. No fractional exponents.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the radical bar is recognised as a grouping symbol by comparing $\sqrt{9 + 16}$ with $\sqrt{9} + \sqrt{16}$, so the rule is derived from the difference rather than announced.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 3 and 6 — students say whether a negative answer is possible at all before they evaluate, which is the whole square-root-versus-cube-root distinction.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits between 13 and 17 on $\sqrt{25 + 144}$, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can evaluate an expression that contains square roots, cube roots and integer exponents.

Essential question. What does the radical bar do to the order in which an expression is evaluated?

Building on. Unit 2 (2.01–2.06) — perfect squares, perfect cubes and the principal square root; 3.04 — negative exponents; Grade 6 — the order of operations.

Working toward. 4.06 — the whole order of operations at once; 4.07–08 — real-world multi-step problems; MA.912.NSO.1.4 in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; this skill continues tomorrow: one assignment, due at the start of the class after 4.06.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is $\sqrt{(9 + 16)}$ not the same as $\sqrt{9} + \sqrt{16}$, when $(9 + 16) \div 5$ is the same as $9 \div 5 + 16 \div 5$?	DOK 3
A cube root of a negative number is ordinary and a square root of one is not. What is it about the exponent 2 that makes the difference?	DOK 3
$\sqrt{49}$ is 7, but the equation $x^2 = 49$ has two solutions. Are those two statements in conflict?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$\sqrt{121}$	***11***	60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]; ±11 — the radical sign names the principal root only [8.NSO.1.7]
2	A square patio has an area of 169 square feet. Fencing for it costs 12 dollars a foot. How long is one side of the patio?	***13 feet***	anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]; 52 — the perimeter, not a side; 84.5 — the area was halved
3	$\sqrt[3]{-125}$	***−5***	5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]; 'not a real number' — the square-root rule applied to a cube root [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]
4	$\sqrt{2^3+17}$	***5***	25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]; anything with two separate roots — the bar groups, so it is one radicand [8.NSO.1.7]
5	√(25 + 144). Which is the value?	***D — 13***	(A) the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]; (B) the sum was found and the root was never taken [8.NSO.1.7]; (C) the radicand was halved instead of rooted [Unit 2, 2.02: a root is not a half]
6	The cube root of −1/64. Which is the value?	***A — −1/4***	(B) the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]; (C) the square root of 1/64 was taken and the sign kept [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]; (D) the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]
7	$\sqrt{\frac{4}{9}}+\sqrt[3]{\frac{8}{27}}$	***4/3***	12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]; 2/3 — only one of the two radicals was evaluated
8	$\sqrt{5^3-25}\cdot 10^{-3}$	***1/100***	10,000 — 10^{-3} read as 10^{3} [8.NSO.1.3: a negative exponent is a reciprocal]; −1/100 — the exponent's sign made the value's sign [B1G-M 8.NSO.1.3 task: −b vs b^{-1}]
9	Simon says that the cube root of −27 is not a real number, because you cannot take the root of a negative. Explain why he is wrong, and give the value.	***−3***	'−3' alone — the value without the reason [the item asks why he is wrong]; '3' — the sign dropped as well as the rule misread

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the perfect-square and perfect-cube lists from Unit 2's reference sheet and let them stay out; a root is a lookup today, not a recall. On every board with arithmetic under the bar, the student draws a box around the radicand before writing anything, so the grouping is a physical act.

ELL. Radicand and radical are the two new nouns, and they are easy to confuse because they sound alike. Write $\sqrt{25}$ once with RADICAL circling the whole symbol and RADICAND labelling only the 25, and point at the labels on every board.

Enrichment. Ask for a pair of numbers for which $\sqrt{(a + b)}$ really does equal $\sqrt{a} + \sqrt{b}$, and what has to be true of them. Then ask why the same question for a product has a different answer.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say which root Simon's rule belongs to, not only that he is wrong, and give −3. Read the boards, not the papers.

Vocabulary. **radical**; **radicand**; **principal square root**

Materials. Whiteboards and markers, and Unit 2's reference sheet if students still want the perfect-square list. Calculators are allowed on everything; a calculator will refuse the cube root of a negative on some models, which is worth showing once and explaining rather than trusting.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, this skill continues tomorrow: one assignment, due at the start of the class after 4.06: Evaluate radical expressions — D8J.

Order of Operations with Radicals

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Evaluate multi-step expressions with rational numbers, exponents and radicals, in six steps or fewer.

Must not No mnemonic. The state guide says to avoid PEMDAS because it leaves out grouping symbols, and the word appears on no student page in this unit.

Target I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

MTR **MTR.5.1** — Notes I · **MTR.3.1** — Boards 1, 3 and 7 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. There is nothing to memorise. Read the expression, ask what is holding what, and the order tells you itself.
2. Nothing says multiply before divide. They are one level, taken left to right like a sentence.
3. A fraction bar is a grouping symbol on the top and another one on the bottom.

Slide by Slide

1. **Order of Operations with Radicals.** Post the learning target.
2. **Warm-Up — On your own. No notes.** Q4 is the prerequisite and it is the one half the room gets wrong. *1 retrieves 4.05 — finish under the bar first: 25 − 9 = 16 (yesterday). 2 retrieves 4.04 — the middle zero counts and the trailing zero counts, because a decimal point is written (last week). 3 retrieves 3.05 — a negative exponent flips the base (last unit). 4 retrieves MA.5.AR.2.2 — multiplication and division are one level, taken left to right (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 4 2. 4 3. 9/4 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the table.** There is nothing to memorise here. Read the expression, ask what is holding what, and the order tells you itself.
6. **Notes.** Nothing in mathematics says multiply before divide. They are the same level, and you read left to right like a sentence.
7. **Notes.** A fraction bar is a grouping symbol on the top and another one on the bottom. Finish each, then divide.
8. **Example 1 — A grouping symbol inside a grouping symbol.** SPOKEN ONLY.
9. **Example 1 — Worked.** The power is inside the parentheses, so it goes first even though it is a level below grouping — because it is inside.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same shape, a square root and a positive answer.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A fraction bar and a radical in one expression.** SPOKEN ONLY.
13. **Example 2 — Worked.** Three groups, each finished on its own before anything crosses between them.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Same three groups, and both halves land on a half.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. $90 \div 5 \cdot 3$ → ***54*** *6 — 5 · 3 was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]; 1,350 — the division was read as a multiplication*

18–19. Board 2. A square tile has side √196 centimeters. Its diagonal is about 19.8 centimeters. What is the perimeter of the tile? → ***56 centimeters*** *The diagonal of about 19.8 centimeters is not needed. anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]; 14 — the side, not the perimeter; 196 — the root was never taken*

20–21. Board 3. $16-5+4$ → ***15*** *7 — 5 + 4 was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]*

22–23. Board 4. $\frac{3^3-21}{5^2-15}$ → ***3/5*** *anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]; −6/5 — 5^2 read as 5 · 2, leaving 6 over −5; 6/25 — the −15 under the bar was dropped*

24–25. Board 5. $2 + 3 \cdot 4^2$. Which is the value? → ***B — 50*** *(A) the exponent was applied to $3 \cdot 4$ instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]; (C) 4^2 was read as $4 \cdot 2$ [8.NSO.1.3: an exponent is not a factor]; (D) 2 + 3 was done first, then $5 \cdot 16$ [8.NSO.1.7: multiplication comes before addition]   Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. $(6^2 - 4)$ over 2^3. Which is the value? → ***A — 4*** *(B) only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]; (C) 6^2 was divided by 2^3 first and then 4 was subtracted [8.NSO.1.7: the bar groups the whole numerator]; (D) the numerator was multiplied by 2^3 instead of divided [8.NSO.1.7: the bar is division]   Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. $-2^2+(-2)^2$ → ***0*** *8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]; −8 — both terms read as -2^2*

30–31. Board 8. $12\left(\left(\frac{1}{2}\right)^3+3^{-2}\right)$ → ***17/6*** *12/8 + 12/9 left unadded — a value, not an answer; 1/8 + 1/9 = 2/17 — the denominators were added [Grade 5: adding fractions needs a common denominator]*

32–33. Board 9. Roscoe says that $(\sqrt{n} - 11)^2$ is always positive, because a square is always positive. Is he right? Give a value of n that settles it. → ***No — at n = 121 the value is 0*** *'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]; 'no' with no value of n — the counterexample is the evidence*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–6 are one structural decision each: multiply before adding, left-to-right division, a power inside parentheses, a fraction bar, the minus-sign-and-exponent contrast, and a radical with a negative exponent. 7 is the three items the state assessment supplies and 8 is three of our own with a bracket, a fraction bar and a cube root. 9 works backwards in three shapes. The pattern breaks at 10, where the student judges a rule rather than a value. 11 is the diagnostic multiple choice, whose three wrong options are three different levels misapplied, and 12 is the select-all. The Additional sheet mirrors each position, and its question 10 is the same error one level down — addition and subtraction rather than multiplication and division.

Misconceptions to Watch

Board 1. 6 — $5 \cdot 3$ was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]

Board 2. anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]

Board 3. 7 — 5 + 4 was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]

Board 4. anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]

Board 5. the exponent was applied to $3 \cdot 4$ instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]

Board 6. only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]

Board 7. 8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]

Board 8. 12/8 + 12/9 left unadded — a value, not an answer

Board 9. 'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]

Materials Whiteboards and markers. Calculators are allowed on everything, and this is the lesson where a calculator is worth using against the class: enter -2^2 and $(-2)^2$ and let the machine settle it.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

Lesson Plan

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1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 6 — Order of Operations with Radicals	Date	
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Evaluate multi-step expressions with rational numbers, exponents and radicals, in six steps or fewer.

Must not. No mnemonic. The state guide says to avoid PEMDAS because it leaves out grouping symbols, and the word appears on no student page in this unit.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the four levels are read off the expression's structure and the class is never given a mnemonic, because the guide asks us not to give one.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1, 3 and 7 — one structural decision each, written and up on the cue; these are the errors that cost marks everywhere else.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits between 50 and 80, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

Essential question. What decides the order in which an expression is evaluated?

Building on. 4.05 — the radical bar as a grouping symbol; 3.04–3.05 — negative exponents on integer and rational bases; Grade 6 — the order of operations with whole numbers.

Working toward. 4.07–08 — the same structure inside real-world problems; MA.912.NSO.1.4 in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 4 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is -2^2 different from $(-2)^2$? What is the exponent attached to in each one?	DOK 3
A mnemonic for the order of operations has no letter for the radical bar or the fraction bar. What does reading the structure give you that the mnemonic cannot?	DOK 3
Roscoe says a square is always positive. What would have to be true about $\sqrt{n} - 11$ for that to be right, and what value of n breaks it?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$90 \div 5 \cdot 3$	***54***	6 — 5 · 3 was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]; 1,350 — the division was read as a multiplication
2	A square tile has side √196 centimeters. Its diagonal is about 19.8 centimeters. What is the perimeter of the tile?	***56 centimeters***	anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]; 14 — the side, not the perimeter; 196 — the root was never taken
3	$16 - 5 + 4$	***15***	7 — 5 + 4 was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]
4	$\frac{3^3 - 21}{5^2 - 15}$	***3/5***	anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]; −6/5 — 5^2 read as 5 · 2, leaving 6 over −5; 6/25 — the −15 under the bar was dropped
5	$2 + 3 \cdot 4^2$. Which is the value?	***B — 50***	(A) the exponent was applied to 3 · 4 instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]; (C) 4^2 was read as 4 · 2 [8.NSO.1.3: an exponent is not a factor]; (D) 2 + 3 was done first, then 5 · 16 [8.NSO.1.7: multiplication comes before addition]
6	$(6^2 - 4)$ over 2^3. Which is the value?	***A — 4***	(B) only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]; (C) 6^2 was divided by 2^3 first and then 4 was subtracted [8.NSO.1.7: the bar groups the whole numerator]; (D) the numerator was multiplied by 2^3 instead of divided [8.NSO.1.7: the bar is division]
7	$-2^2 + (-2)^2$	***0***	8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]; −8 — both terms read as -2^2
8	$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$	***17/6***	12/8 + 12/9 left unadded — a value, not an answer; 1/8 + 1/9 = 2/17 — the denominators were added [Grade 5: adding fractions needs a common denominator]
9	Roscoe says that $(\sqrt{n} - 11)^2$ is always positive, because a square is always positive. Is he right? Give a value of n that settles it.	***No — at n = 121 the value is 0***	'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]; 'no' with no value of n — the counterexample is the evidence

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the four-level table from Notes I as a card and let it stay out. On every expression the student first circles all the grouping symbols and numbers them from the inside out, and only then evaluates; the circling is the work, and the arithmetic follows it.

ELL. Grouping symbol is the phrase to own. Put the four of them on the board — parentheses, brackets, the radical bar, the fraction bar — and have the student point to each one in an expression and name it before evaluating anything.

Enrichment. Ask for an expression using 2, 3 and 4 once each, with any grouping symbols, whose value is as large as possible — then as small as possible. Then ask why a mnemonic could not have answered either question.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to give a value of n that settles Roscoe's claim and say what is wrong with the sentence, not only that it is wrong. Read the boards, not the papers.

Vocabulary. **grouping symbol**; **nested**; **multi-step expression**

Materials. Whiteboards and markers. Calculators are allowed on everything, and this is the lesson where a calculator is worth using against the class: enter -2^2 and $(-2)^2$ and let the machine settle it.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 4 test: Evaluate radical expressions — D8J.

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 7–8

Real-World Order of Operations

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Solve multi-step real-world problems with the order of operations, exponents and radicals, in six steps or fewer, and report the answer with its unit.

Must not No keyword strategy and no CUBES — the state guide names circling numbers and boxing key words as a misconception, not a method. No PEMDAS.

Target I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

MTR **MTR.7.1** — Notes I–III, both Examples and boards 2, 3, 4, 6, 7 and 9 · **MTR.2.1** — Notes II and board 9 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Every substituted value goes into parentheses. That one habit prevents half of what goes wrong today.
2. There is no word that tells you which operation to use. Read what is happening, not which words are there.
3. Round once, at the very end, and write the unit. An answer with no unit is not an answer.

Slide by Slide

1. Real-World Order of Operations. Post the learning target.

2. Warm-Up — On your own. No notes. Q4 is the prerequisite and it carries the day's one habit. *1 retrieves 4.06 — the exponent attaches to its own base; the parentheses decide what that base is (yesterday). 2 retrieves 4.05 — finish each piece, then combine: 12 + 1/2 (last week). 3 retrieves 3.05 — a negative exponent flips the base (last unit). 4 retrieves MA.6.AR.1.3 — substitute, then follow the structure: $2(3)^2 = 2(9)$ (prior grade — the prerequisite this lesson needs).*

3. Warm-Up — Answers. Reveal. 1. 0 2. 25/2 3. 4/3 4. 18

4. Vocabulary — The new words, before they are used. Read each word and its meaning once; the notes use them next.

5. Notes — Copy both lines. Every substituted value goes into parentheses. That single habit prevents half of what goes wrong in this lesson.

6. Notes — Copy both problems. There is no word that tells you which operation to use. Read what is happening, not which words are present.

7. Notes. Round once, at the very end. Rounding in the middle moves the answer, and it moves it further every step you take afterwards.

8. Example 1 — A formula with a power in it. SPOKEN ONLY.

9. Example 1 — Worked. Substitute in parentheses, square the speed, then multiply.

10. Your Turn — Same steps, your numbers. Boards up when done. Same formula, smaller numbers.

11. Your Turn — Answer. Reveal; name what a wrong board most likely did (see the note above).

12. Example 2 — A cost with a rate and a flat fee. SPOKEN ONLY.

13. Example 2 — Worked. The two per-square-foot prices go inside the group that is multiplied by the area; the flat fee stays outside it.

14. Your Turn — Same steps, your numbers. Boards up when done. One rate and one flat fee this time.

15. Your Turn — Answer. Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. Evaluate $3x^2$ when $x = -2$. → ***12*** *−12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]; 36 — the exponent was applied to 3x [8.NSO.1.7: the exponent belongs to its own base]*

18–19. Board 2. A cyclist has a mass of 70 kilograms and rides at 8 meters per second. Her bicycle cost 480 dollars. Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v. Find her kinetic energy. → ***2,240 joules*** *The price of the bicycle is not needed. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]; 560 — v^2 read as 2v [8.NSO.1.3: an exponent is not a factor]; 280 — the square dropped altogether; 2,240 with no unit — the answer is in joules*

20–21. Board 3. Renting a kayak costs 17 dollars plus 9 dollars for each hour. Find the cost of 3 hours. → ***44 dollars*** *78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]; 29 — 17 + 9 + 3, every number added [B1G-M 8.NSO.1.7 misconception: keyword hunting]*

22–23. Board 4. A square floor of side 8 feet is tiled at 3.50 dollars a square foot. Installation is a flat 60 dollars. Find the total cost. → ***284 dollars*** *4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]; 88 — 8(3.50) + 60: the side used instead of the area*

24–25. Board 5. Evaluate ½mv^2 when m = 40 and v = 3. Which is the value? → ***A — 180*** *(B) v^2 was read as 2v [8.NSO.1.3: an exponent is not a factor]; (C) the one half was dropped [8.NSO.1.7: every factor in the formula is used]; (D) the exponent was applied to the whole product, $(½ \cdot 40 \cdot 3)^2$ [8.NSO.1.7: the exponent belongs to the v alone] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A square patio of side 12 feet is sealed at 1.25 dollars a square foot. Delivery of the sealer is a flat 38 dollars. Which expression gives the total cost? → ***D — $12^2(1.25) + 38$*** *(A) the side was used where the area is needed [8.NSO.1.7: read what the rate is per]; (B) the prices were added to the area instead of multiplied by it [B1G-M 8.NSO.1.7 misconception: keyword hunting]; (C) the flat delivery charge was put inside the area group, so it is charged on every square foot [8.NSO.1.7: a flat fee is not a rate] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A square garden has 4 sides, each √150 feet long. Find its perimeter, to the nearest hundredth of a foot. → ***48.99 feet*** *48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]; 48.99 with no unit — the answer is in feet*

30–31. Board 8. Evaluate $(a^2 − b) ÷ c$ when a = 7, b = 4, c = 5. → ***9*** *48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]; 5.8 — 49 ÷ 5 done first, then 4 subtracted*

32–33. Board 9. One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings. Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery. Find both totals, and say why the key words could not have told you. → ***23 dollars and 31 dollars*** *23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]; 23 twice — the second structure was not read*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–4 substitute into four shapes: a coefficient with a power, a formula with two letters, a fraction bar, and a binomial inside a power. 5 and 6 are cost expressions — one square, one rectangle — where the student has to decide what multiplies the area. 7 and 8 are two real formulas, each evaluated three times, so the structure is fixed and only the numbers move; 7 also requires rounding to a stated place. 9 works backwards in three shapes, ending with a cost equation that has to be unwound. The pattern breaks at 10, where two problems with the same numbers and the same words have different answers. 11 is the diagnostic multiple choice on cost structure and 12 the select-all on what an exponent is attached to. The Additional sheet mirrors each position.

Misconceptions to Watch

Board 1. −12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]

Board 2. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]

Board 3. 78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]

Board 4. 4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]

Board 5. v^2 was read as 2v [8.NSO.1.3: an exponent is not a factor]

Board 6. the side was used where the area is needed [8.NSO.1.7: read what the rate is per]

Board 7. 48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]

Board 8. 48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]

Board 9. 23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]

Materials Whiteboards and markers. Calculators are allowed on everything and are the right tool for the BMI and kinetic-energy items; the work being assessed is the expression, not the arithmetic.

Real-World Order of Operations

Lesson Plan

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1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lessons 7–8 — Real-World Order of Operations	Date	________________
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Solve multi-step real-world problems with the order of operations, exponents and radicals, in six steps or fewer, and report the answer with its unit.

Must not. No keyword strategy and no CUBES — the state guide names circling numbers and boxing key words as a misconception, not a method. No PEMDAS.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Notes I–III, both Examples and boards 2, 3, 4, 6, 7 and 9 — the expression arrives inside a situation, and the unit is part of the answer.
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II and board 9 — the same three numbers are written as two different expressions, so the structure is seen to come from the situation and not from the page.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits over whether the flat fee goes inside the parentheses, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

Essential question. If key words cannot tell you which operation a problem needs, what can?

Building on. 4.05–4.06 — grouping symbols and the four levels; MA.6.AR.1 — substituting a value for a letter; Unit 2 — rounding at the end, not in the middle.

Working toward. The Unit 4 assessment; MA.8.AR.2 and MA.912.NSO.1.4, where formulas are rearranged rather than only evaluated.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the unit test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two problems have the same numbers and the same words and different answers. What is it in the problem, then, that decides the operation?	DOK 3
Why does rounding $\sqrt{200}$ to 14 before multiplying by 6 lose most of a foot, when 14 is only about 0.14 away from $\sqrt{200}$?	DOK 3
In a cost expression, what has to be true of a price for it to belong inside the parentheses that are multiplied by the area?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Evaluate $3x^2$ when $x=-2$.	***12***	−12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]; 36 — the exponent was applied to 3x [8.NSO.1.7: the exponent belongs to its own base]
2	A cyclist has a mass of 70 kilograms and rides at 8 meters per second. Her bicycle cost 480 dollars. Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v. Find her kinetic energy.	***2,240 joules***	anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]; 560 — v^2 read as 2v [8.NSO.1.3: an exponent is not a factor]; 280 — the square dropped altogether; 2,240 with no unit — the answer is in joules
3	Renting a kayak costs 17 dollars plus 9 dollars for each hour. Find the cost of 3 hours.	***44 dollars***	78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]; 29 — 17 + 9 + 3, every number added [B1G-M 8.NSO.1.7 misconception: keyword hunting]
4	A square floor of side 8 feet is tiled at 3.50 dollars a square foot. Installation is a flat 60 dollars. Find the total cost.	***284 dollars***	4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]; 88 — 8(3.50) + 60: the side used instead of the area
5	Evaluate $\frac{1}{2}mv^2$ when m = 40 and v = 3. Which is the value?	***A — 180***	(B) v^2 was read as 2v [8.NSO.1.3: an exponent is not a factor]; (C) the one half was dropped [8.NSO.1.7: every factor in the formula is used]; (D) the exponent was applied to the whole product, $(\frac{1}{2} \cdot 40 \cdot 3)^2$ [8.NSO.1.7: the exponent belongs to the v alone]
6	A square patio of side 12 feet is sealed at 1.25 dollars a square foot. Delivery of the sealer is a flat 38 dollars. Which expression gives the total cost?	***D — $12^2(1.25) + 38$***	(A) the side was used where the area is needed [8.NSO.1.7: read what the rate is per]; (B) the prices were added to the area instead of multiplied by it [B1G-M 8.NSO.1.7 misconception: keyword hunting]; (C) the flat delivery charge was put inside the area group, so it is charged on every square foot [8.NSO.1.7: a flat fee is not a rate]
7	A square garden has 4 sides, each √150 feet long. Find its perimeter, to the nearest hundredth of a foot.	***48.99 feet***	48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]; 48.99 with no unit — the answer is in feet
8	Evaluate $(a^2 - b) \div c$ when a = 7, b = 4, c = 5.	***9***	48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]; 5.8 — 49 ÷ 5 done first, then 4 subtracted
9	One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings. Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery. Find both totals, and say why the key words could not have told you.	***23 dollars and 31 dollars***	23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]; 23 twice — the second structure was not read

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give a two-step frame — WRITE THE EXPRESSION, then EVALUATE IT — and require the expression on paper before any arithmetic. Substituted values go in parentheses every time, with the parentheses pre-printed on the frame for the first two problems.

ELL. Rate and flat fee are the two phrases every cost problem turns on, and 'a square foot' is what marks the difference. Put 'dollars a square foot' and 'a flat 60 dollars' side by side on the board and ask which one depends on the size of the floor, before any expression is written.

Enrichment. Ask for a third problem using 15, 4 and 6 whose answer is neither 39 nor 66. Then ask whether every arrangement of three numbers and two operations can be made into a sensible situation.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to give both totals and say what decides the operation, since the words and the numbers are identical in the two problems. Read the boards, not the papers.

Vocabulary. **formula**; **substitute**; **unit**

Materials. Whiteboards and markers. Calculators are allowed on everything and are the right tool for the BMI and kinetic-energy items; the work being assessed is the expression, not the arithmetic.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the unit test: Evaluate variable expressions: word problems — DPG.

GRADE 7 ACCELERATED · UNIT 4

Unit 4 Review

Solving Problems with Rational Numbers

Name______________________

Date______________________

Period____________________

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers each part of the assessment, with more questions than the assessment has. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

Adding and Subtracting in Scientific Notation

Lesson 4.01 · 6 questions

1. $(4.1 \times 10^{6}) + (2.3 \times 10^{6})$

2. $(7.6 \times 10^{-4}) - (2.9 \times 10^{-4})$

3. $(8.2 \times 10^{5}) + (4.9 \times 10^{5})$

4. $(5.4 \times 10^{8}) + (7.1 \times 10^{7})$

5. $(6.3 \times 10^{-2}) - (8 \times 10^{-4})$

6. Lena writes $(3.2\times10^4)+(5.5\times10^6)=8.7\times10^{10}$. Name her two mistakes and give the correct sum.

Multiplying and Dividing in Scientific Notation

Lessons 4.02–4.03 · 6 questions

7. $(4\times10^3)(2\times10^6)$

8. $\frac{7.2\times10^8}{1.8\times10^3}$

9. $(5\times10^6)(8\times10^{-2})$

10. $\frac{2.4\times10^5}{8\times10^9}$

11. $\frac{4\times10^7}{4\times10^{-3}}$

12. A laser emits 6.5×10^{14} photons each second. How many does it emit in 4×10^{2} seconds?

Real-World Problems in Scientific Notation

Lesson 4.04 · 4 questions

13. A rectangular panel measures 3.2×10^{2} mm by 1.45×10^{2} mm. Find its area, in scientific notation. Do not round.

14. A train travels 4.8×10^{2} km in 6 hours. Find its average speed, in scientific notation. Do not round.

15. Two lengths are measured: 5.4×10^{3} mm and 2.8×10^{2} mm. Find their sum, in scientific notation. Do not round.

16. Two masses are measured: 7.6×10^{-2} g and 3.42×10^{-3} g. Find their difference, in scientific notation. Do not round.

17. $\sqrt{169}$

18. $\sqrt[3]{-8}$

19. $\sqrt{7^2 - 24}$

20. $\sqrt[3]{\frac{1}{27}}$

21. $\sqrt{100} \cdot 4^{-2}$

22. $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2 + 7}$

23. Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

The Order of Operations

Lesson 4.06 · 7 questions

24. $24 - 3 \cdot 5 + 1$

25. $48 \div 6 \cdot 2$

26. $4 + 3(8 - 5)^2$

27. $\frac{6^2 + 9}{3^2 - 4}$

28. $-6^2 + (-6)^2$

29. $20\left(\left(\frac{1}{2}\right)^2 + 5^{-1}\right)$

30. Which expression is equivalent in value to $2+4\cdot5^2$?

- ○ A. 900
- ○ B. 102
- ○ C. 42
- ○ D. 150

Real-World Problems and the Order of Operations

Lessons 4.07–4.08 · 5 questions

31. Evaluate $5x^2$ when $x=-3$.

32. Kinetic energy is $K=\frac{1}{2}mv^2$. Find K when $m=400$ kg and $v=7$ m/s.

33. A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

34. Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery.

a. What is the area of the patio?

b. Write an expression for the total cost.

c. Find the total cost.

d. A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

35. A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

○ A. $18^2(3.50+140)$
○ B. $18(3.50)+140$
○ C. $18^2+3.50+140$
○ D. $18^2(3.50)+140$

GRADE 7 ACCELERATED · UNIT 4

Unit 4 Review

Solving Problems with Rational Numbers **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers each part of the assessment, with more questions than the assessment has. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

Adding and Subtracting in Scientific Notation

Lesson 4.01 · MA.8.NSO.1.5 · 6 questions

1. $(4.1\times10^{6})+(2.3\times10^{6})$

6.4×10^{6}

Same power: add the coefficients, copy the power down.

2. $(7.6\times10^{-4})-(2.9\times10^{-4})$

4.7×10^{-4}

3. $(8.2\times10^{5})+(4.9\times10^{5})$

1.31×10^{6}

13.1×10^{5} *is the value and not the finished form.*

4. $(5.4\times10^{8})+(7.1\times10^{7})$

6.11×10^{8}

$7.1\times10^{7}=0.71\times10^{8}$.

5. $(6.3\times10^{-2})-(8\times10^{-4})$

6.22×10^{-2}

$8\times10^{-4}=0.08\times10^{-2}$ *; the exponents are two apart, the widest gap allowed.*

6. Lena writes $(3.2\times10^{4})+(5.5\times10^{6})=8.7\times10^{10}$. Name her two mistakes and give the correct sum.

She added the coefficients without first writing both over the same power of ten, and she added the exponents. The sum is 5.532×10^{6} **.**

$3.2\times10^{4}=0.032\times10^{6}$ *, and* $5.5+0.032=5.532$.

Multiplying and Dividing in Scientific Notation

Lessons 4.02–4.03 · MA.8.NSO.1.5 · 6 questions

7. $(4\times10^3)(2\times10^6)$

8×10^9

Multiply the coefficients, add the exponents.

8. $\frac{7.2\times10^8}{1.8\times10^3}$

4×10^5

Divide the coefficients, subtract the exponents, top minus bottom.

9. $(5\times10^6)(8\times10^{-2})$

4×10^5

40×10^4 *is not finished.*

10. $\frac{2.4\times10^5}{8\times10^9}$

3×10^{-5}

0.3×10^{-4} *is not finished either.*

11. $\frac{4\times10^7}{4\times10^{-3}}$

1×10^{10}

$7-(-3)=10$ *. Subtracting a negative goes up.*

12. A laser emits 6.5×10^{14} photons each second. How many does it emit in 4×10^2 seconds?

2.6×10^{17} ***photons***

$6.5\cdot4=26$ *, so* 26×10^{16} *before it is finished.*

Real-World Problems in Scientific Notation

Lesson 4.04 · MA.8.NSO.1.6 · 4 questions

13. A rectangular panel measures 3.2×10^2 mm by 1.45×10^2 mm. Find its area, in scientific notation. Do not round.

4.64×10^4 ***square millimeters***

$3.2\cdot1.45=4.64$ *, and* $10^2\cdot10^2=10^4$.

14. A train travels 4.8×10^{2} km in 6 hours. Find its average speed, in scientific notation. Do not round.

8×10^{1} ***km/h***

$\frac{4.8\times10^{2}}{6}=0.8\times10^{2}$ *, which is not finished.*

15. Two lengths are measured: 5.4×10^{3} mm and 2.8×10^{2} mm. Find their sum, in scientific notation. Do not round.

5.68×10^{3} ***mm***

$2.8\times10^{2}=0.28\times10^{3}$ *, and* $5.4+0.28=5.68$ *.*

16. Two masses are measured: 7.6×10^{-2} g and 3.42×10^{-3} g. Find their difference, in scientific notation. Do not round.

7.258×10^{-2} ***g***

$3.42\times10^{-3}=0.342\times10^{-2}$ *, and* $7.6-0.342=7.258$ *.*

Evaluating Expressions with Radicals

Lesson 4.05 · MA.8.NSO.1.7 · 7 questions

17. $\sqrt{169}$

13

18. $\sqrt[3]{-8}$

−2

An odd root keeps the sign of its radicand.

19. $\sqrt{7^{2}-24}$

5

Finish under the bar: 49 − 24 = 25.

20. $\sqrt[3]{\frac{1}{27}}$

$\frac{1}{3}$

Top and bottom each get the root.

21. $\sqrt{100}\cdot4^{-2}$

$\frac{5}{8}$

$10\cdot\frac{1}{16}$.

22. $\left(-\frac{1}{2}\right)^2 + \sqrt{3^2 + 7}$

$\frac{17}{4}$

$\frac{1}{4} + 4$.

23. Kai says that $\sqrt[3]{-64}$ is not a real number. Is he right? Explain.

No. A cube root asks for the number used three times, and $(-4)(-4)(-4) = -64$ ***, so the value is −4. It is a square root of a negative number that is not a real number.***

This is the first misconception the state guide names for this benchmark.

The Order of Operations

Lesson 4.06 · MA.8.NSO.1.7 · 7 questions

24. $24 - 3 \cdot 5 + 1$

10

Multiply first, then left to right.

25. $48 \div 6 \cdot 2$

16

One level, left to right: 8 · 2. Not 4.

26. $4 + 3(8 - 5)^2$

31

$4 + 3(9)$.

27. $\frac{6^2 + 9}{3^2 - 4}$

9

45 over 5.

28. $-6^2 + (-6)^2$

0

The exponent attaches to its own base; the parentheses decide what that base is.

29. $20\left(\left(\frac{1}{2}\right)^2 + 5^{-1}\right)$

9

$20 \cdot \frac{9}{20}$.

30. Which expression is equivalent in value to $2+4\cdot5^2$?

- ○ A. 900
- ● **B.** 102
- ○ C. 42
- ○ D. 150

B

Real-World Problems and the Order of Operations *Lessons 4.07–4.08 · MA.8.NSO.1.7 · 5 questions*

31. Evaluate $5x^2$ when $x=-3$.

45

$5(-3)^2=5(9)$.

32. Kinetic energy is $K=\frac{1}{2}mv^2$. Find K when $m=400$ kg and $v=7$ m/s.

9,800 joules

$200(49)$ *. The unit is part of the answer.*

33. A gym charges 30 dollars to join plus 12 dollars for each of 4 months. A second gym charges 30 dollars for each of 12 months plus 4 dollars to join. Find both totals.

78 dollars and 364 dollars

$30+12(4)=78$ *and* $30(12)+4=364$ *. The same three numbers and the same words; what decides is which amount is repeated.*

34. **SSDD.** Every part below is about a square patio of side 15 feet, sealed at 2.20 dollars a square foot with a flat 60 dollars for delivery. Same surface, four different questions: a measurement, an expression, an evaluation, and a size read back out of a total. Only part b has no number for an answer, and only part d runs the structure backwards.

a. What is the area of the patio?

225 square feet

b. Write an expression for the total cost.

$15^2(2.20)+60$

The rate multiplies the area; the flat fee does not.

c. Find the total cost.

555 dollars

$225(2.20)=495$ *, then* $+60$.

d. A second square patio, at the same rate and the same flat fee, costs 940 dollars. What is its side?

20 feet

$940-60=880$ *, and* $880\div2.20=400$ *, whose square root is 20.*

35. A square court of side 18 feet is resurfaced at 3.50 dollars a square foot, with a flat 140 dollars for the equipment. Which expression gives the total cost?

- ○ A. $18^2(3.50+140)$
- ○ B. $18(3.50)+140$
- ○ C. $18^2+3.50+140$
- ● **D.** $18^2(3.50)+140$

D

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points

Name___________________
Date___________________
Period_________________

Show your work. One point for each lettered part.

Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ○ C. 2.914×10^5
- ○ D. 4.3×10^8

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☐ A. $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☐ C. $(6 \times 10^3)(5 \times 10^2)$
- ☐ D. $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☐ F. $\frac{9 \times 10^9}{3 \times 10^3}$

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1} . What was the number? Write your answer in scientific notation. *1 pt*

Real-World Problems in Scientific Notation

4 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^{2} mm by 1.35×10^{2} mm. Find its area. *1 pt*

b. A vehicle travels 5.6×10^{2} km in 8 hours. Find its average speed. *1 pt*

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^{3} mm and 4.1×10^{2} mm. Find their sum. *1 pt*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

The Order of Operations with Exponents and Radicals

13 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

b. $\frac{5^2 + 7}{2^3}$ *1 pt*

c. $36 \div 4 \cdot 3$ *1 pt*

d. $-4^2 + (-4)^2$ *1 pt*

10. Evaluate each expression.

a. $\sqrt{6^2 + 13}$ *1 pt*

b. $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

c. $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

b. Find the total cost. *1 pt*

12. Which is the value of $2 + 5 \cdot 3^2$? *1 pt*

- ○ A. 32
- ○ B. 47
- ○ C. 63
- ○ D. 289

13. Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☐ B. $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☐ D. $\left(\frac{1}{6}\right)^{-1}$
- ☐ E. $\sqrt{5^2 + 11}$
- ☐ F. $\frac{4^2 + 8}{4}$

14. Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Operations in Scientific Notation

12 points MA.8.NSO.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

5.7×10^5

Same power: add the coefficients.

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

2.5×10^{-6}

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

1.43×10^9

14.3×10^8 *is the value; 14.3 is not less than 10, so it is not the answer.*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

4.57×10^4

$3.7 \times 10^3 = 0.37 \times 10^4$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

6×10^{10}

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

8×10^5

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

2×10^5

20×10^4 *is not finished.*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3×10^{-4}

0.3×10^{-3} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ● C. 2.914×10^5
- ○ D. 4.3×10^8

C

Scoring: 1 point.

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☒ **A.** $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☒ **C.** $(6 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☒ **F.** $\frac{9 \times 10^9}{3 \times 10^3}$

A, C, D and F

C gives 30×10^5 *, which is* 3×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 6.123×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item. $2.3 \times 10^4 = 0.023 \times 10^6$ *.*

Scoring: 1 point.

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1} . What was the number? Write your answer in scientific notation. *1 pt*

5×10^{-5}

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. $\frac{2 \times 10^{-1}}{4 \times 10^3} = 0.5 \times 10^{-4} = 5 \times 10^{-5}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.5×10^{-4} *alone does not earn it.*

Scoring: 1 point.

Real-World Problems in Scientific Notation

4 points MA.8.NSO.1.6

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^2 mm by 1.35×10^2 mm. Find its area. *1 pt*

3.24×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 2.4 · 1.35 = 3.24, and $10^2 \cdot 10^2 = 10^4$ *.*

b. A vehicle travels 5.6×10^2 km in 8 hours. Find its average speed. *1 pt*

7×10^1 ***km/h***

$\frac{5.6 \times 10^2}{8} = 0.7 \times 10^2$ *, which is not finished: 0.7 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^3 mm and 4.1×10^2 mm. Find their sum. *1 pt*

6.66×10^3 ***mm***

Match the powers first: $4.1 \times 10^2 = 0.41 \times 10^3$ *, and 6.25 + 0.41 = 6.66.*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

8.828×10^{-2} ***g***

Match the powers first: $4.72 \times 10^{-3} = 0.472 \times 10^{-2}$ *, and 9.3 − 0.472 = 8.828.*

Scoring: a 1 · b 1 = 2 points.

The Order of Operations with Exponents and Radicals

13 points MA.8.NSO.1.7

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

−3

The power first, then the product, then the subtraction.

b. $\frac{5^2 + 7}{2^3}$ *1 pt*

4

32 over 8; the bar groups the whole numerator.

c. $36 \div 4 \cdot 3$ *1 pt*

27

One level, left to right: 9 · 3. Not 3.

d. $-4^2 + (-4)^2$ *1 pt*

0

−16 + 16. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{6^2+13}$ *1 pt*

7

Finish under the bar: 36 + 13 = 49.

b. $\sqrt[3]{-125}+2^{-2}$ *1 pt*

$-\frac{19}{4}$

$-5+\frac{1}{4}$.

c. $\left(-\frac{1}{3}\right)^2+\sqrt{\frac{4}{9}}$ *1 pt*

$\frac{7}{9}$

$\frac{1}{9}+\frac{2}{3}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$14^2(2.40)+85$

The rate multiplies the area; the flat fee does not. An expression that charges 85 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

555.40 dollars

196(2.40) = 470.40, then + 85.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $2+5\cdot3^2$? *1 pt*

- ○ A. 32
- ● **B. 47**
- ○ C. 63
- ○ D. 289

B

Scoring: 1 point.

13. Select ALL of the expressions whose value is 6. *1 pt*

☐ A. $-\sqrt{36}$

☒ **B.** $\sqrt{36}$

☐ C. $36 \div 3 \cdot 2$

☒ **D.** $\left(\frac{1}{6}\right)^{-1}$

☒ **E.** $\sqrt{5^2 + 11}$

☒ **F.** $\frac{4^2 + 8}{4}$

B, D, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 25 + 144 is finished before the root is taken: $\sqrt{169} = 13$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '13' alone does not earn it, and neither does 'she added wrong'.

Scoring: 1 point.

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

676 square inches

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The side of the first is $\sqrt{169} = 13$ *, so the second has side 26 and area* $26^2 = 676$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 338 has not gone through the side.*

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.NSO.1.5	1, 2, 3, 4, 5, 6	12	
MA.8.NSO.1.6	7, 8	4	
MA.8.NSO.1.7	9, 10, 11, 12, 13, 14, 15	13	
Total		**29**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is *arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form A · 29 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

5.7×10^5

Same power: add the coefficients.

b. $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

2.5×10^{-6}

c. $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

1.43×10^9

14.3×10^8 *is the value; 14.3 is not less than 10, so it is not the answer.*

d. $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

4.57×10^4

$3.7 \times 10^3 = 0.37 \times 10^4$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

6×10^{10}

b. $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

8×10^5

c. $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

2×10^5

20×10^4 *is not finished.*

d. $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

3×10^{-4}

0.3×10^{-3} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$? *1 pt*

- ○ A. 3.04×10^5
- ○ B. 4.3×10^5
- ● C. 2.914×10^5
- ○ D. 4.3×10^8

C

Scoring: 1 point.

4. Select ALL of the expressions whose value is 3×10^6 . *1 pt*

- ☒ **A.** $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☒ **C.** $(6 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☒ **F.** $\frac{9 \times 10^9}{3 \times 10^3}$

A, C, D and F

C gives 30×10^5 *, which is* 3×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 6.123×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item. $2.3 \times 10^4 = 0.023 \times 10^6$ *.*

Scoring: 1 point.

6. A number is multiplied by 4×10^3 and the product is 2×10^{-1}. What was the number? Write your answer in scientific notation. *1 pt*

5×10^{-5}

$\frac{2 \times 10^{-1}}{4 \times 10^3} = 0.5 \times 10^{-4} = 5 \times 10^{-5}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.5×10^{-4} *alone does not earn it.*

Scoring: 1 point.

Real-World Problems in Scientific Notation

4 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 2.4×10^2 mm by 1.35×10^2 mm. Find its area. *1 pt*

3.24×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 2.4 · 1.35 = 3.24, and $10^2 \cdot 10^2 = 10^4$ *.*

b. A vehicle travels 5.6×10^2 km in 8 hours. Find its average speed. *1 pt*

7×10^1 ***km/h***

$\frac{5.6 \times 10^2}{8} = 0.7 \times 10^2$ *, which is not finished: 0.7 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 6.25×10^3 mm and 4.1×10^2 mm. Find their sum. *1 pt*

6.66×10^3 ***mm***

Match the powers first: $4.1 \times 10^2 = 0.41 \times 10^3$ *, and 6.25 + 0.41 = 6.66.*

b. Two masses are measured: 9.3×10^{-2} g and 4.72×10^{-3} g. Find their difference. *1 pt*

8.828×10^{-2} ***g***

Match the powers first: $4.72 \times 10^{-3} = 0.472 \times 10^{-2}$ *, and 9.3 − 0.472 = 8.828.*

Scoring: a 1 · b 1 = 2 points.

The Order of Operations with Exponents and Radicals

13 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $15 - 2 \cdot 3^2$ *1 pt*

−3

The power first, then the product, then the subtraction.

b. $\frac{5^2 + 7}{2^3}$ *1 pt*

4

32 over 8; the bar groups the whole numerator.

c. $36 \div 4 \cdot 3$ *1 pt*

27

One level, left to right: 9 · 3. Not 3.

d. $-4^2 + (-4)^2$ *1 pt*

0

−16 + 16. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{6^2+13}$ *1 pt*

7

Finish under the bar: 36 + 13 = 49.

b. $\sqrt[3]{-125}+2^{-2}$ *1 pt*

$-\frac{19}{4}$

$-5+\frac{1}{4}$.

c. $\left(-\frac{1}{3}\right)^2+\sqrt{\frac{4}{9}}$ *1 pt*

$\frac{7}{9}$

$\frac{1}{9}+\frac{2}{3}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$14^2(2.40)+85$

The rate multiplies the area; the flat fee does not. An expression that charges 85 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

555.40 dollars

196(2.40) = 470.40, then + 85.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $2+5\cdot3^2$? *1 pt*

- ○ A. 32
- ● **B. 47**
- ○ C. 63
- ○ D. 289

B

Scoring: 1 point.

13. Select ALL of the expressions whose value is 6. *1 pt*

☐ A. $-\sqrt{36}$

☒ **B.** $\sqrt{36}$

☐ C. $36 \div 3 \cdot 2$

☒ **D.** $\left(\frac{1}{6}\right)^{-1}$

☒ **E.** $\sqrt{5^2 + 11}$

☒ **F.** $\frac{4^2+8}{4}$

B, D, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Nadia writes $\sqrt{25+144} = \sqrt{25} + \sqrt{144} = 17$. Explain why she is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 25 + 144 is finished before the root is taken: $\sqrt{169} = 13$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '13' alone does not earn it, and neither does 'she added wrong'.

Scoring: 1 point.

15. A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

676 square inches

The side of the first is $\sqrt{169} = 13$ *, so the second has side 26 and area* $26^2 = 676$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 338 has not gone through the side.*

Scoring: 1 point.

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points

Name___________________
Date___________________
Period_________________

Show your work. One point for each lettered part.

Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^{6}) + (2.4 \times 10^{6})$ *1 pt*

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

c. $(6.4 \times 10^{7}) + (8.2 \times 10^{7})$ *1 pt*

d. $(5.3 \times 10^{5}) + (2.6 \times 10^{4})$ *1 pt*

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

- ○ A. 6.4×10^5
- ○ B. 4.123×10^5
- ○ C. 4.33×10^5
- ○ D. 6.4×10^8

4. Select ALL of the expressions whose value is 4×10^6. *1 pt*

☐ A. $(2.5 \times 10^6) + (1.5 \times 10^6)$
☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
☐ C. $(8 \times 10^3)(5 \times 10^2)$
☐ D. $(4.3 \times 10^6) - (3 \times 10^5)$
☐ E. $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2}. What was the number? Write your answer in scientific notation. *1 pt*

Real-World Problems in Scientific Notation

4 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

The Order of Operations with Exponents and Radicals

13 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

b. $\frac{4^2 + 9}{5}$ *1 pt*

c. $45 \div 9 \cdot 3$ *1 pt*

d. $-3^2 - (-3)^2$ *1 pt*

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

b. Find the total cost. *1 pt*

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ○ D. 39

13. Select ALL of the expressions whose value is 8. *1 pt*

- ☐ A. $\sqrt{64}$
- ☐ B. $64 \div 4 \cdot 2$
- ☐ C. $\left(\frac{1}{8}\right)^{-1}$
- ☐ D. $-\sqrt{64}$
- ☐ E. $\frac{5^2 + 7}{4}$
- ☐ F. $\sqrt{6^2 + 28}$

14. Tomas writes $\sqrt{36 + 64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Operations in Scientific Notation

12 points MA.8.NSO.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^6) + (2.4 \times 10^6)$ *1 pt*

6.7×10^6

Same power: add the coefficients.

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

5.5×10^{-5}

c. $(6.4 \times 10^7) + (8.2 \times 10^7)$ *1 pt*

1.46×10^8

14.6×10^7 *is the value; 14.6 is not less than 10, so it is not the answer.*

d. $(5.3 \times 10^5) + (2.6 \times 10^4)$ *1 pt*

5.56×10^5

$2.6 \times 10^4 = 0.26 \times 10^5$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

8×10^8

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

4×10^5

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

3×10^3

30×10^2 *is not finished.*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3×10^{-5}

0.3×10^{-4} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

○ A. 6.4×10^5

● **B.** 4.123×10^5

○ C. 4.33×10^5

○ D. 6.4×10^8

B

Scoring: 1 point.

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

☒ **A.** $(2.5 \times 10^6) + (1.5 \times 10^6)$

☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$

☒ **C.** $(8 \times 10^3)(5 \times 10^2)$

☒ **D.** $(4.3 \times 10^6) - (3 \times 10^5)$

☒ **E.** $\frac{1.2 \times 10^{10}}{3 \times 10^3}$

☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

A, C, D and E

C gives 40×10^5 *, which is* 4×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 5.234×10^6 ***.***

Both mistakes must be named for the point; one of the two is half the item. $3.4 \times 10^4 = 0.034 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2}. What was the number? Write your answer in scientific notation. *1 pt*

6×10^{-7}

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. $\frac{3 \times 10^{-2}}{5 \times 10^4} = 0.6 \times 10^{-6} = 6 \times 10^{-7}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.6×10^{-6} *alone does not earn it.*

Scoring: 1 point.

Real-World Problems in Scientific Notation

4 points MA.8.NSO.1.6

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

4.375×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 3.5 · 1.25 = 4.375, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

9×10^1 ***km/h***

$\frac{7.2 \times 10^2}{8} = 0.9 \times 10^2$ *, which is not finished: 0.9 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

7.87×10^3 ***mm***

Match the powers first: $5.2 \times 10^2 = 0.52 \times 10^3$ *, and 7.35 + 0.52 = 7.87.*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

8.039×10^{-2} ***g***

Match the powers first: $3.61 \times 10^{-3} = 0.361 \times 10^{-2}$ *, and 8.4 − 0.361 = 8.039.*

Scoring: a 1 · b 1 = 2 points.

The Order of Operations with Exponents and Radicals *13 points MA.8.NSO.1.7*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

8

The power first, then the product, then the subtraction.

b. $\frac{4^2 + 9}{5}$ *1 pt*

5

25 over 5; the bar groups the whole numerator.

c. $45 \div 9 \cdot 3$ *1 pt*

15

One level, left to right: 5 · 3. Not 5.

d. $-3^2 - (-3)^2$ *1 pt*

−18

−9 − 9. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

9

Finish under the bar: 64 + 17 = 81.

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

$-\frac{35}{9}$

$-4 + \frac{1}{9}$.

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

1

$\frac{1}{4} + \frac{3}{4}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$12^2(2.60) + 75$

The rate multiplies the area; the flat fee does not. An expression that charges 75 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

449.40 dollars

144(2.60) = 374.40, then + 75.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ● **D. 39**

D

Scoring: 1 point.

13. Select ALL of the expressions whose value is 8. *1 pt*

☒ **A.** $\sqrt{64}$

☐ B. $64 \div 4 \cdot 2$

☒ **C.** $\left(\frac{1}{8}\right)^{-1}$

☐ D. $-\sqrt{64}$

☒ **E.** $\frac{5^2+7}{4}$

☒ **F.** $\sqrt{6^2+28}$

A, C, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 36 + 64 is finished before the root is taken: $\sqrt{100} = 10$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '10' alone does not earn it, and neither does 'he added wrong'.

Scoring: 1 point.

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

900 square inches

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The side of the first is $\sqrt{225} = 15$ *, so the second has side 30 and area* $30^2 = 900$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 450 has not gone through the side.*

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.NSO.1.5	1, 2, 3, 4, 5, 6	12	
MA.8.NSO.1.6	7, 8	4	
MA.8.NSO.1.7	9, 10, 11, 12, 13, 14, 15	13	
Total		**29**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is *arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*

GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

Solving Problems with Rational Numbers

Form B · 29 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Operations in Scientific Notation

12 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. Add or subtract. Write each answer in scientific notation.

a. $(4.3 \times 10^6) + (2.4 \times 10^6)$ *1 pt*

6.7×10^6

Same power: add the coefficients.

b. $(9.1 \times 10^{-5}) - (3.6 \times 10^{-5})$ *1 pt*

5.5×10^{-5}

c. $(6.4 \times 10^7) + (8.2 \times 10^7)$ *1 pt*

1.46×10^8

14.6×10^7 *is the value; 14.6 is not less than 10, so it is not the answer.*

d. $(5.3 \times 10^5) + (2.6 \times 10^4)$ *1 pt*

5.56×10^5

$2.6 \times 10^4 = 0.26 \times 10^5$.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

2. Multiply or divide. Write each answer in scientific notation.

a. $(4 \times 10^3)(2 \times 10^5)$ *1 pt*

8×10^8

b. $\frac{8.4 \times 10^8}{2.1 \times 10^3}$ *1 pt*

4×10^5

c. $(6 \times 10^{-4})(5 \times 10^6)$ *1 pt*

3×10^3

30×10^2 *is not finished.*

d. $\frac{1.5 \times 10^2}{5 \times 10^6}$ *1 pt*

3×10^{-5}

0.3×10^{-4} *is not finished either.*

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

3. Which is the value of $(2.3 \times 10^3) + (4.1 \times 10^5)$? *1 pt*

○ A. 6.4×10^5
● **B.** 4.123×10^5
○ C. 4.33×10^5
○ D. 6.4×10^8

B

Scoring: 1 point.

4. Select ALL of the expressions whose value is 4×10^6 . *1 pt*

☒ **A.** $(2.5 \times 10^6) + (1.5 \times 10^6)$
☐ B. $\frac{1.2 \times 10^{10}}{3 \times 10^4}$
☒ **C.** $(8 \times 10^3)(5 \times 10^2)$
☒ **D.** $(4.3 \times 10^6) - (3 \times 10^5)$
☒ **E.** $\frac{1.2 \times 10^{10}}{3 \times 10^3}$
☐ F. $(2.5 \times 10^6) + (1.5 \times 10^5)$

A, C, D and E

C gives 40×10^5 *, which is* 4×10^6 *once it is finished. All four correct and nothing else for the point.*

Scoring: 1 point.

5. Devon writes $(3.4 \times 10^4) + (5.2 \times 10^6) = 8.6 \times 10^{10}$. Name the two mistakes he made, and give the correct sum. *1 pt*

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 5.234×10^6***.***

Both mistakes must be named for the point; one of the two is half the item. $3.4 \times 10^4 = 0.034 \times 10^6$.

Scoring: 1 point.

6. A number is multiplied by 5×10^4 and the product is 3×10^{-2}. What was the number? Write your answer in scientific notation. *1 pt*

6×10^{-7}

$\frac{3 \times 10^{-2}}{5 \times 10^4} = 0.6 \times 10^{-6} = 6 \times 10^{-7}$. *Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* 0.6×10^{-6} *alone does not earn it.*

Scoring: 1 point.

Real-World Problems in Scientific Notation *4 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

7. Solve. Write each answer in scientific notation. Do not round.

a. A rectangular plate measures 3.5×10^2 mm by 1.25×10^2 mm. Find its area. *1 pt*

4.375×10^4 ***square millimeters***

Multiply the coefficients and add the exponents: 3.5 · 1.25 = 4.375, and $10^2 \cdot 10^2 = 10^4$.

b. A vehicle travels 7.2×10^2 km in 8 hours. Find its average speed. *1 pt*

9×10^1 ***km/h***

$\frac{7.2 \times 10^2}{8} = 0.9 \times 10^2$*, which is not finished: 0.9 is less than 1.*

Scoring: a 1 · b 1 = 2 points.

8. Solve. Write each answer in scientific notation. Do not round.

a. Two lengths are measured: 7.35×10^3 mm and 5.2×10^2 mm. Find their sum. *1 pt*

7.87×10^3 ***mm***

Match the powers first: $5.2 \times 10^2 = 0.52 \times 10^3$ *, and 7.35 + 0.52 = 7.87.*

b. Two masses are measured: 8.4×10^{-2} g and 3.61×10^{-3} g. Find their difference. *1 pt*

8.039×10^{-2} ***g***

Match the powers first: $3.61 \times 10^{-3} = 0.361 \times 10^{-2}$ *, and 8.4 − 0.361 = 8.039.*

Scoring: a 1 · b 1 = 2 points.

The Order of Operations with Exponents and Radicals *13 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. Evaluate each expression.

a. $20 - 3 \cdot 2^2$ *1 pt*

8

The power first, then the product, then the subtraction.

b. $\frac{4^2 + 9}{5}$ *1 pt*

5

25 over 5; the bar groups the whole numerator.

c. $45 \div 9 \cdot 3$ *1 pt*

15

One level, left to right: 5 · 3. Not 5.

d. $-3^2 - (-3)^2$ *1 pt*

−18

−9 − 9. The parentheses decide what the exponent is attached to.

Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.

10. Evaluate each expression.

a. $\sqrt{8^2 + 17}$ *1 pt*

9

Finish under the bar: 64 + 17 = 81.

b. $\sqrt[3]{-64} + 3^{-2}$ *1 pt*

$-\frac{35}{9}$

$-4 + \frac{1}{9}$.

c. $\left(-\frac{1}{2}\right)^2 + \sqrt{\frac{9}{16}}$ *1 pt*

1

$\frac{1}{4} + \frac{3}{4}$.

Scoring: a 1 · b 1 · c 1 = 3 points.

11. A square patio of side 12 feet is sealed at 2.60 dollars a square foot, with a flat 75 dollars for delivery.

a. Write an expression for the total cost. *1 pt*

$12^2(2.60) + 75$

The rate multiplies the area; the flat fee does not. An expression that charges 75 on every square foot does not earn the point.

b. Find the total cost. *1 pt*

449.40 dollars

144(2.60) = 374.40, then + 75.

Scoring: a 1 · b 1 = 2 points.

12. Which is the value of $3 + 4 \cdot 3^2$? *1 pt*

- ○ A. 27
- ○ B. 63
- ○ C. 225
- ● **D. 39**

D

Scoring: 1 point.

13. Select ALL of the expressions whose value is 8. *1 pt*

☒ **A.** $\sqrt{64}$

☐ B. $64 \div 4 \cdot 2$

☒ **C.** $\left(\frac{1}{8}\right)^{-1}$

☐ D. $-\sqrt{64}$

☒ **E.** $\frac{5^2+7}{4}$

☒ **F.** $\sqrt{6^2+28}$

A, C, E and F

All four correct and nothing else for the point.

Scoring: 1 point.

14. Tomas writes $\sqrt{36+64} = \sqrt{36} + \sqrt{64} = 14$. Explain why he is wrong, and give the correct value. *1 pt*

The radical bar is a grouping symbol, so 36 + 64 is finished before the root is taken: $\sqrt{100} = 10$ ***. The root of a sum is not the sum of the roots.***

The point needs the reason AND the value. '10' alone does not earn it, and neither does 'he added wrong'.

Scoring: 1 point.

15. A square tile has an area of 225 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

900 square inches

The side of the first is $\sqrt{225} = 15$ *, so the second has side 30 and area* $30^2 = 900$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 450 has not gone through the side.*

Scoring: 1 point.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

Independent Set 4.01 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $(4.1\times10^{6})+(2.7\times10^{6})$

6.8×10^{6}

2. $(9.3\times10^{-5})-(5.1\times10^{-5})$

4.2×10^{-5}

3. $(7.5\times10^{3})+(8.9\times10^{3})$

1.64×10^{4}

The sum is 16.4×10^{3} *, which is not finished.*

4. $(6.2\times10^{7})+(3.5\times10^{6})$

6.55×10^{7}

$3.5\times10^{6}=0.35\times10^{7}$.

5. $(4.8\times10^{-2})-(9.0\times10^{-4})$

4.71×10^{-2}

$9.0\times10^{-4}=0.09\times10^{-2}$.

6. Find n . $(5.2\times10^{9})+(n\times10^{9})=1.31\times10^{10}$

n = 7.9

$1.31\times10^{10}=13.1\times10^{9}$.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

Question Bank 4.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(2.4 \times 10^5) + (3.1 \times 10^5)$

5.5×10^5

2. $(8.3 \times 10^{-4}) - (2.1 \times 10^{-4})$

6.2×10^{-4}

3. $(6.7 \times 10^9) + (5.8 \times 10^9)$

$12.5 \times 10^9 = 1.25 \times 10^{10}$

The sum of the coefficients is over 10, so the answer needs one more step.

4. $(9.1 \times 10^4) - (8.6 \times 10^4)$

$0.5 \times 10^4 = 5 \times 10^3$

A coefficient below 1 is unfinished too: point one place right, exponent down one.

5. $(7.2 \times 10^6) + (4.3 \times 10^5)$

7.63×10^6

$4.3 \times 10^5 = 0.43 \times 10^6$ *, and* $7.2 + 0.43 = 7.63$.

6. $(5.5 \times 10^{-3}) - (8.0 \times 10^{-5})$

5.42×10^{-3}

$8.0 \times 10^{-5} = 0.08 \times 10^{-3}$ *. The exponents are two apart, the widest gap allowed.*

Real quantities.

7. Neptune's diameter is about 4.95×10^4 kilometers. Saturn's is about 1.21×10^5 kilometers.

a. What is the sum of the two diameters?

1.705×10^5 ***km***

$1.21 \times 10^5 = 12.1 \times 10^4$ *, so the sum is* 17.05×10^4 .

b. Jupiter's diameter is about 1.43×10^5 kilometers. Is your answer to part (a) greater than Jupiter's diameter?

Yes — $1.705 \times 10^5 > 1.43 \times 10^5$ **.**

Same power of ten, so the coefficients decide.

8. A city's population is about 4.62×10^5 . The surrounding county's population is about 1.38×10^6 .

a. What is the total population of the city and the county?

1.842×10^6 ***people***

$4.62 \times 10^5 = 0.462 \times 10^6$.

b. How many more people live in the county than in the city?

9.18×10^5 ***people***

$1.38 \times 10^6 = 13.8 \times 10^5$ *, and* $13.8 - 4.62 = 9.18$.

Working backwards.

9. Find the missing number in each equation.

a. $(3.6 \times 10^7) + (n \times 10^7) = 1.01 \times 10^8$

n = 6.5

$1.01 \times 10^8 = 10.1 \times 10^7$.

b. $(8.4 \times 10^{-6}) - (n \times 10^{-6}) = 2.9 \times 10^{-6}$

n = 5.5

c. $(2.5 \times 10^4) + (7.5 \times 10^4) =$ ______

1×10^5

The coefficients sum to exactly 10, so the answer is 10×10^4 *— finished, that is* 1×10^5 .

Reading someone else's work.

10. Priya writes $(2.6\times10^4)+(5.1\times10^6)=7.7\times10^{10}$. Name the two mistakes she made, and give the correct sum.

She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is 5.126×10^6 **.**

$2.6\times10^4=0.026\times10^6$ *, and* $5.1+0.026=5.126$.

11. Which is the value of $(4.7\times10^{-5})+(8.6\times10^{-5})$, written in scientific notation?

- ○ A. 13.3×10^{-10}
- ● **B.** 1.33×10^{-4}
- ○ C. 1.33×10^{-10}
- ○ D. 13.3×10^{-5}

B

12. Select ALL of the expressions whose value is 1.2×10^5 .

- ☒ **A.** $(1.25\times10^5)-(5\times10^3)$
- ☐ B. $(1.3\times10^5)-(1\times10^5)$
- ☒ **C.** $(1.3\times10^5)-(1\times10^4)$
- ☐ D. $(8\times10^4)+(4\times10^5)$
- ☒ **E.** $(8\times10^4)+(4\times10^4)$
- ☒ **F.** $(1.1\times10^5)+(1\times10^4)$

A, C, E and F

E gives 12×10^4 *, which is* 1.2×10^5 *once it is finished.*

GRADE 7 ACCELERATED · UNIT 4 · LESSON 1

Adding and Subtracting in Scientific Notation

Question Bank - Additional 4.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(5.3\times10^{7})+(2.4\times10^{7})$

7.7×10^{7}

2. $(9.6\times10^{-3})-(4.2\times10^{-3})$

5.4×10^{-3}

3. $(7.4\times10^{5})+(8.9\times10^{5})$

$16.3\times10^{5}=1.63\times10^{6}$

4. $(6.3\times10^{-7})-(5.9\times10^{-7})$

$0.4\times10^{-7}=4\times10^{-8}$

Point one place right, exponent down one.

5. $(3.8\times10^{9})+(6.1\times10^{8})$

4.41×10^{9}

$6.1\times10^{8}=0.61\times10^{9}$.

6. $(2.7\times10^{-2})-(9.0\times10^{-4})$

2.61×10^{-2}

$9.0\times10^{-4}=0.09\times10^{-2}$.

Real quantities.

7. Mars's diameter is about 6.79×10^{3} kilometers. Earth's is about 1.28×10^{4} kilometers.

a. What is the sum of the two diameters?

1.959×10^{4} ***km***

$6.79\times10^{3}=0.679\times10^{4}$.

b. How much larger is Earth's diameter than Mars's?

6.01×10^{3} ***km***

$1.28\times10^{4}=12.8\times10^{3}$ *, and* $12.8-6.79=6.01$.

8. A charity raised about 4.85×10^5 dollars in its first year and about 1.27×10^6 dollars in its second.

a. How much did it raise in the two years together?

1.755×10^6 ***dollars***

$4.85\times10^5=0.485\times10^6$.

b. How much more did it raise in the second year than in the first?

7.85×10^5 ***dollars***

$1.27\times10^6=12.7\times10^5$.

Working backwards.

9. Find the missing number in each equation.

a. $(5.2\times10^8)+(n\times10^8)=1.34\times10^9$

n = 8.2

$1.34\times10^9=13.4\times10^8$.

b. $(9.6\times10^{-4})-(n\times10^{-4})=3.8\times10^{-4}$

n = 5.8

c. $(6.5\times10^3)+(3.5\times10^3)=$ ______

1×10^4

The coefficients sum to exactly 10.

Reading someone else's work.

10. Malik writes $(3.2\times10^5)+(7.1\times10^7)=1.03\times10^{13}$. Name the two mistakes he made, and give the correct sum.

He added the coefficients without first writing both numbers over the same power of ten, and he added the exponents. The correct sum is 7.132×10^7 **.**

$3.2\times10^5=0.032\times10^7$ *, and* $7.1+0.032=7.132$ *. His* 10.3×10^{12} *was then repaired correctly to* 1.03×10^{13} *— the repair is the one thing he did right.*

11. Which is the value of $(6.8\times10^{-6})+(5.4\times10^{-6})$, written in scientific notation?

● A. 1.22×10^{-5}
○ B. 12.2×10^{-12}
○ C. 12.2×10^{-6}
○ D. 1.22×10^{-12}

A

12. Select ALL of the expressions whose value is 2.4×10^6 .

☒ **A.** $(1.2 \times 10^6) + (1.2 \times 10^6)$

☐ B. $(1.8 \times 10^6) + (6 \times 10^6)$

☒ **C.** $(2.42 \times 10^6) - (2 \times 10^4)$

☒ **D.** $(1.8 \times 10^6) + (6 \times 10^5)$

☒ **E.** $(2.5 \times 10^6) - (1 \times 10^5)$

☐ F. $(2.5 \times 10^6) - (1 \times 10^6)$

A, C, D and E

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

Independent Set 4.02 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $(2\times10^{5})(4\times10^{3})$

8×10^{8}

2. $\frac{9.9\times10^{7}}{3.3\times10^{2}}$

3×10^{5}

3. $(5\times10^{6})(6\times10^{-2})$

3×10^{5}

The product is 30×10^{4} *, which is not finished.*

4. $\frac{1.8\times10^{3}}{6\times10^{7}}$

3×10^{-5}

The quotient is 0.3×10^{-4} *, which is not finished either.*

5. $\frac{4\times10^{9}}{8\times10^{-2}}$

5×10^{10}

$9-(-2)=11$ *, and* 0.5×10^{11} *is not finished.*

6. Find n . $(2\times10^{n})(3\times10^{4})=6\times10^{10}$

n = 6

$n+4=10$.

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

Question Bank 4.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(2\times 10^{3})(4\times 10^{5})$

8×10^{8}

2. $\frac{8.8\times 10^{9}}{2.2\times 10^{4}}$

4×10^{5}

3. $(6\times 10^{4})(5\times 10^{7})$

$30\times 10^{11}=3\times 10^{12}$

The coefficients multiply to 30, so the answer needs one more step.

4. $\frac{1.5\times 10^{6}}{5\times 10^{9}}$

$0.3\times 10^{-3}=3\times 10^{-4}$

A coefficient below 1 is unfinished: point one place right, exponent down one.

5. $(3.5\times 10^{-6})(4\times 10^{-3})$

$14\times 10^{-9}=1.4\times 10^{-8}$

Both exponents are negative, so the product law adds them to −9.

6. $\frac{4.8\times 10^{-2}}{1.6\times 10^{-7}}$

3×10^{5}

$-2-(-7)=5$

Real quantities.

7. Three meteorites have masses 1.1×10^2 , 6.8×10^2 and 8.4×10^{-2} grams.

a. How many times as heavy as the lightest one is the heaviest one? Round to the nearest whole number.

about 8,095 times

$680 \div 0.084 = 8{,}095.2\ldots$

b. How many times as heavy as the lightest one is the 1.1×10^2 gram one? Round to the nearest whole number.

about 1,310 times

$110 \div 0.084 = 1{,}309.5\ldots$

8. A data center processes about 4.5×10^6 records per hour.

a. How many records does it process in 2.4×10^2 hours?

1.08×10^9 ***records***

$4.5 \cdot 2.4 = 10.8$ *, so the product is* 10.8×10^8 *before it is finished.*

b. A second center processed 9.0×10^9 records in 1.8×10^3 hours. How many records per hour is that?

5×10^6 ***records per hour***

Working backwards.

9. Find the missing number in each equation.

a. $(3 \times 10^n)(2 \times 10^4) = 6 \times 10^{11}$

n = 7

$n + 4 = 11$.

b. $\frac{9.6 \times 10^8}{n \times 10^3} = 1.2 \times 10^5$

n = 8

$9.6 \div n = 1.2$.

c. $(5 \times 10^4)(4 \times 10^6) =$ ______

2×10^{11}

The coefficients multiply to 20, so 20×10^{10} *becomes* 2×10^{11} .

Reading someone else's work.

10. Tyree multiplies $(2.8\times10^3)(9.7\times10^6)$ and writes 27.16×10^9. Is his value correct? Is his answer written in scientific notation? Write the finished answer.

The value is correct. It is not in scientific notation, because 27.16 is not less than 10. Finished: 2.716×10^{10} **.**

$2.8\cdot9.7=27.16$ *and* $10^3\cdot10^6=10^9$ *, so every step he took was right — he stopped one step early.*

11. Which is the value of $\frac{7.5\times10^{-4}}{2.5\times10^2}$?

- ○ A. 5×10^{-6}
- ● **B.** 3×10^{-6}
- ○ C. 3×10^{6}
- ○ D. 3×10^{-2}

B

12. Select ALL of the expressions whose value is 6×10^8 .

- ☒ **A.** $(1.5\times10^4)(4\times10^4)$
- ☒ **B.** $\frac{3\times10^{10}}{5\times10^1}$
- ☒ **C.** $(2\times10^3)(3\times10^5)$
- ☐ D. $\frac{1.2\times10^{12}}{2\times10^4}$
- ☐ E. $(2\times10^3)(3\times10^{15})$
- ☒ **F.** $\frac{1.2\times10^{12}}{2\times10^3}$

A, B, C and F

F and B each give 0.6×10^9 *, which is* 6×10^8 *once it is finished.*

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 2–3

Multiplying and Dividing in Scientific Notation

Question Bank - Additional 4.02 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, multiply or divide. Write each answer in scientific notation.

1. $(3\times10^2)(3\times10^6)$

9×10^8

2. $\frac{6.3\times10^7}{2.1\times10^3}$

3×10^4

3. $(4\times10^5)(8\times10^6)$

$32\times10^{11}=3.2\times10^{12}$

4. $\frac{2.4\times10^4}{8\times10^7}$

$0.3\times10^{-3}=3\times10^{-4}$

5. $(2.5\times10^{-5})(6\times10^{-4})$

$15\times10^{-9}=1.5\times10^{-8}$

6. $\frac{9.9\times10^{-3}}{3.3\times10^{-8}}$

3×10^5

$-3-(-8)=5$.

Real quantities.

7. The Amazon River discharges about 5.5×10^7 gallons of water every second.

a. About how many gallons does it discharge in one hour, which is 3.6×10^3 seconds?

about 1.98×10^{11} ***gallons***

$5.5\cdot3.6=19.8$ *, so the product is* 19.8×10^{10} *before it is finished.*

b. A second river discharges about 1.1×10^7 gallons per second. How many times larger is the Amazon's rate?

5 times larger

8. A factory produces about 6.4×10^5 bottles per day.

a. How many bottles does it produce in 3.5×10^2 days?

2.24×10^8 ***bottles***

$6.4\cdot3.5=22.4$.

b. A second factory produced 7.2×10^8 bottles in 1.2×10^3 days. How many bottles per day is that?

6×10^5 ***bottles per day***

Working backwards.

9. Find the missing number in each equation.

a. $(4\times10^n)(2\times10^3)=8\times10^{10}$

n = 7

$n+3=10$.

b. $\frac{7.2\times10^6}{n\times10^2}=2.4\times10^4$

n = 3

$7.2\div n=2.4$.

c. $(2.5\times10^3)(8\times10^5)=$ ______

2×10^9

The coefficients multiply to exactly 20.

Reading someone else's work.

10. Dana divides $\frac{4.8\times10^{11}}{1.2\times10^4}$ and writes 3.6×10^7 . Name her mistake and give the correct quotient.

She subtracted the coefficients instead of dividing them. The power of ten is right. The correct quotient is 4×10^7 **.**

$4.8-1.2=3.6$ *, which is where her coefficient came from;* $4.8\div1.2=4$.

11. Which is the value of $\frac{9.6\times10^{-3}}{3.2\times10^5}$?

- ● **A.** 3×10^{-8}
- ○ B. 3×10^8
- ○ C. 6.4×10^{-8}
- ○ D. 3×10^2

A

12. Select ALL of the expressions whose value is 8×10^6 .

☒ **A.** $(1.6 \times 10^3)(5 \times 10^3)$

☒ **B.** $\frac{1.6 \times 10^{10}}{2 \times 10^3}$

☐ C. $(4 \times 10^2)(2 \times 10^8)$

☒ **D.** $\frac{4 \times 10^8}{5 \times 10^1}$

☒ **E.** $(4 \times 10^2)(2 \times 10^4)$

☐ F. $\frac{1.6 \times 10^{10}}{2 \times 10^4}$

A, B, D and E

B and D each give 0.8×10^7 *, which is* 8×10^6 *once it is finished.*

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Independent Set 4.04 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. How many significant digits are in 6,030 grams?

3

The middle zero counts; the trailing zero does not, because no decimal point is written.

2. How many significant digits are in 0.00710 meters?

3

Leading zeros never count; the trailing zero does, because a decimal point is written.

3. A rectangle measures 2.4×10^2 m by 3.15×10^2 m. Find its area, to the correct number of significant digits.

7.6×10^4 ***square meters***

75,600 exactly, kept to two significant digits.

4. $(6.30 \times 10^4) + (2.8 \times 10^3)$ meters, both measured. Write the sum with the correct precision.

6.58×10^4 ***m***

63,000 and 2,800 both stop at the hundreds place, so the sum, 65,800, does too.

5. A car travels 1.80×10^2 km in 4.0 hours. Find its average speed, to the correct number of significant digits.

4.5×10^1 ***km/h***

45 exactly; two significant digits.

6. $(3.7 \times 10^{-2}) - (8.45 \times 10^{-4})$ grams, both measured. Write the difference with the correct precision.

3.6×10^{-2} ***g***

0.036155 exactly; 0.037 stops at the thousandths place, so the answer stops there.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Question Bank 4.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, give the number of significant digits in each measurement.

1. 4,207 grams

4

Every digit counts: the zero sits between two nonzero digits.

2. 0.0058 meters

2

Leading zeros are never significant.

3. 3.200 seconds

4

The trailing zeros count, because a decimal point is written. 3.200 and 3.2 are the same number but not the same claim.

4. 52,000 litres

2

No decimal point is written, so the trailing zeros are only holding places.

For questions 5 through 8, solve. Report each answer with the correct precision.

5. A rectangular plot measures 3.5×10^2 m by 1.24×10^2 m. Find its area.

4.3×10^4 ***square meters***

43,400 exactly; 2 significant digits and 3, so the answer keeps 2.

6. A car travels 2.60×10^2 km in 3.0 hours. Find its average speed.

8.7×10^1 ***km/h***

260 ÷ 3.0 = 86.66…; 3 significant digits and 2, so the answer keeps 2.

7. Two lengths are measured: 8.7×10^2 mm and 4.5×10^2 mm. Find their sum.

1.32×10^3 ***mm***

870 and 450 both stop at the tens place, so the sum, 1,320, does too. A sum can carry more significant digits than either measurement — three here, from two twos.

8. Two masses are measured: 8.40×10^{-2} g and 3.715×10^{-3} g. Find their difference.

8.03×10^{-2} *g*

0.080285 exactly; 0.0840 stops at the ten-thousandths place and 0.003715 stops two places further out, so the answer stops at the ten-thousandths.

Real quantities.

9. The Amazon River discharges about 5.5×10^{7} gallons of water each second.

a. How many gallons does it discharge in one hour, which is 3.6×10^{3} seconds? Report the answer with the correct number of significant digits.

2.0×10^{11} ***gallons***

1.98×10^{11} *exactly. The rate carries two significant digits; 3,600 seconds in an hour is an exact conversion, not a measurement, so it does not limit the count. The answer keeps two.*

b. A second river discharges about 1.10×10^{7} gallons per second. How many times larger is the Amazon's rate? Report the answer with the correct number of significant digits.

5.0 times larger

Exactly 5; two significant digits are justified, so it is written 5.0.

10. A processor performs about 4.66×10^{8} calculations each second.

a. How many calculations does it perform in one minute? Report the answer with the correct number of significant digits.

2.80×10^{10} ***calculations***

$4.66 \times 10^{8} \cdot 60 = 2.796 \times 10^{10}$ *. Sixty seconds in a minute is exact, not a measurement, so it does not limit the count — the answer keeps the measurement's three digits.*

b. Five of these processors run for 20 minutes. How many calculations do they perform altogether? Report the answer with the correct number of significant digits.

2.80×10^{12} ***calculations***

$2.796 \times 10^{10} \cdot 5 \cdot 20 = 2.796 \times 10^{12}$ *. Five and twenty are counts, so again only the measurement's three digits survive.*

Reading someone else's work.

11. Omar multiplies a measured length of 2.5×10^{2} m by a measured width of 1.346×10^{2} m and reports the area as 3.365×10^{4} square meters. Is his arithmetic correct? Is his answer correct? Explain.

His arithmetic is correct — the exact product is 3.365×10^{4} ***. His answer is not, because 2.5 carries only two significant digits, so the area can carry only two:*** 3.4×10^{4} ***square meters.***

A correct value in the wrong form is still the wrong answer here; the rule is part of the question.

12. A rectangle's sides are measured as 7.2×10^1 cm and 4.0×10^1 cm. Which is the area, in scientific notation with the correct number of significant digits?

- ○ A. 2.9×10^2 cm^2
- ○ B. 28.8×10^2 cm^2
- ● **C.** 2.9×10^3 **cm^2**
- ○ D. 2.88×10^3 cm^2

C

GRADE 7 ACCELERATED · UNIT 4 · LESSON 4

Real-World Problems and Significant Digits

Question Bank - Additional 4.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, give the number of significant digits in each measurement.

1. 9,304 kilograms

4

The zero sits between two nonzero digits.

2. 0.00062 meters

2

Leading zeros are never significant.

3. 1.500 seconds

4

The trailing zeros count because a decimal point is written.

4. 430,000 litres

2

No decimal point, so the trailing zeros only hold places.

For questions 5 through 8, solve. Report each answer with the correct precision.

5. A rectangular field measures 6.5×10^2 m by 2.34×10^2 m. Find its area.

1.5×10^5 ***square meters***

152,100 exactly; the answer keeps 2 significant digits.

6. A train travels 3.40×10^2 km in 4.0 hours. Find its average speed.

8.5×10^1 ***km/h***

340 ÷ 4.0 = 85 exactly; the answer keeps 2 significant digits.

7. Two lengths are measured: 6.28×10^4 mm and 9.1×10^3 mm. Find their sum.

7.19×10^4 ***mm***

62,800 stops at the hundreds place and 9,100 stops at the hundreds place, so the sum, 71,900, stops there too.

8. Two masses are measured: 5.70×10^{-1} g and 2.845×10^{-2} g. Find their difference.

5.42×10^{-1} ***g***

0.54155 exactly; 0.570 stops at the thousandths place, so the answer stops there.

Real quantities.

9. A wind turbine generates about 2.4×10^{3} kilowatt-hours of electricity each hour.

a. How much does it generate in 7.5×10^{3} hours? Report the answer with the correct number of significant digits.

1.8×10^{7} ***kilowatt-hours***

$18\times10^{6}=1.8\times10^{7}$ *exactly; both measurements carry two significant digits.*

b. A second turbine generates about 8.0×10^{2} kilowatt-hours each hour. How many times larger is the first turbine's rate? Report the answer with the correct number of significant digits.

3.0 times larger

Exactly 3; two significant digits are justified, so it is written 3.0.

10. A sensor records about 8.25×10^{5} readings each second.

a. How many readings does it record in one minute? Report the answer with the correct number of significant digits.

4.95×10^{7} ***readings***

$8.25\times10^{5}\cdot60=4.95\times10^{7}$ *exactly. Sixty is a count, not a measurement, so the three digits survive.*

b. Four of these sensors run for 30 minutes. How many readings do they record altogether? Report the answer with the correct number of significant digits.

5.94×10^{9} ***readings***

$4.95\times10^{7}\cdot4\cdot30=5.94\times10^{9}$ *exactly. Four and thirty are counts.*

Reading someone else's work.

11. Bianca adds two measured lengths, 86.4 cm and 0.271 cm. She says that because 86.4 has three significant digits and 0.271 has three, the sum has three: 86.7 cm. Is she right? Explain, and give the correct sum.

She is not right, even though her answer happens to be close. The rule for a sum is about places, not counts: 86.4 stops at the tenths and 0.271 stops at the thousandths, so the sum stops at the tenths. $86.4+0.271=86.671$ ***, which becomes 86.7 cm.***

Here the two rules give the same answer, which is exactly why the wrong rule survives so long. Ask what she would have written for 1,234 + 5.6.

12. A rectangle's sides are measured as 5.0×10^1 cm and 6.4×10^1 cm. Which is the area, in scientific notation with the correct number of significant digits?

- ○ A. 3.2×10^2 cm^2
- ○ B. 3.20×10^3 cm^2
- ○ C. 32×10^2 cm^2
- ● **D.** 3.2×10^3 **cm^2**

D

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Independent Set 4.05 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $\sqrt{225}$

15

2. $\sqrt[3]{-1}$

−1

An odd root keeps the sign.

3. $\sqrt{6^2+13}$

7

36 + 13 = 49.

4. $\sqrt[3]{\frac{1}{8}}$

$\frac{1}{2}$

5. $\sqrt{144}\cdot 2^{-3}$

$\frac{3}{2}$

$12\cdot\frac{1}{8}$.

6. $\left(-\frac{1}{2}\right)^3+\sqrt{4^2+9}$

$\frac{39}{8}$

$-\frac{1}{8}+5$.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Question Bank 4.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{81}$

9

2. $\sqrt[3]{125}$

5

3. $\sqrt[3]{-8}$

−2

An odd root keeps the sign of its radicand.

4. $\sqrt{\frac{4}{25}}$

$\frac{2}{5}$

The numerator and the denominator each get the root.

5. $\sqrt{5^2+11}$

6

Finish under the bar: 25 + 11 = 36.

6. $\sqrt[3]{3^3+37}$

4

27 + 37 = 64.

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{100} \cdot 2^{-2}$

$\frac{5}{2}$

$10 \cdot \frac{1}{4}$.

b. $\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$

$\frac{17}{4}$

$\frac{1}{4} + 4$.

c. $\sqrt[3]{-64} - 3^{-1}$

$-\frac{13}{3}$

$-4 - \frac{1}{3}$.

8. Evaluate each expression.

a. $\sqrt[3]{27} - 1.4\left(\sqrt{3^2 - 5}\right)$

0.2

$3 - 1.4(2)$.

b. $\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$

$-\frac{17}{9}$

$\frac{1}{9} - 2$.

c. $\sqrt{4^3 - 39} \cdot 5^{-3}$

$\frac{1}{25}$

$5 \cdot \frac{1}{125}$.

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n}=12$

n = 144

b. $\sqrt[3]{n}=-5$

n = −125

c. $\sqrt{n+7}=9$

n = 74

$n+7=81$.

Reading someone else's work.

10. Tonya writes $\sqrt{16+9}=\sqrt{16}+\sqrt{9}=7$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that her method ignores.

$\sqrt{16+9}=\sqrt{25}=5$ ***. The bar groups 16 + 9 into a single radicand, so the addition is finished before any root is taken.***

Her method would need the root of a sum to equal the sum of the roots, and it does not.

11. Which is the value of $\sqrt{36+64}$?

○ A. 14
○ B. 100
● **C. 10**
○ D. 50

C

12. Select ALL of the expressions whose value is 3.

☐ A. $\sqrt{3^2+16}$
☒ **B.** $\sqrt[3]{2^2+23}$
☒ **C.** $\sqrt{4^2-7}$
☒ **D.** $\sqrt[3]{27}$
☐ E. $\sqrt[3]{-27}$
☒ **F.** $\sqrt{9}$

B, C, D and F

C's radicand is $16-7=9$ *and B's is* $4+23=27$.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Question Bank - Additional 4.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{64}$

8

2. $\sqrt[3]{8}$

2

3. $\sqrt[3]{-125}$

−5

An odd root keeps the sign of its radicand.

4. $\sqrt{\frac{9}{49}}$

$\frac{3}{7}$

Top and bottom each get the root.

5. $\sqrt{4^2+33}$

7

16 + 33 = 49.

6. $\sqrt[3]{5^2+100}$

5

25 + 100 = 125.

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{144} \cdot 3^{-2}$

$\frac{4}{3}$

$12 \cdot \frac{1}{9}$.

b. $\left(-\frac{1}{4}\right)^2 + \sqrt{3^2 + 27}$

$\frac{97}{16}$

$\frac{1}{16} + 6$.

c. $\sqrt[3]{-27} - 2^{-1}$

$-\frac{7}{2}$

$-3 - \frac{1}{2}$.

8. Evaluate each expression.

a. $\sqrt[3]{64} - 1.5\left(\sqrt{7^2 - 13}\right)$

−5

$4 - 1.5(6)$.

b. $\left(-\frac{1}{2}\right)^3 - \sqrt[3]{4^2 + 11}$

$-\frac{25}{8}$

$-\frac{1}{8} - 3$.

c. $\sqrt{6^2 + 13} \cdot 2^{-3}$

$\frac{7}{8}$

$7 \cdot \frac{1}{8}$.

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 15$

n = 225

b. $\sqrt[3]{n} = -2$

n = −8

c. $\sqrt{n-5} = 11$

n = 126

$n - 5 = 121$.

Reading someone else's work.

10. Deshawn writes $\sqrt{100-36} = \sqrt{100} - \sqrt{36} = 4$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that his method ignores.

$\sqrt{100-36} = \sqrt{64} = 8$ ***. The bar groups the subtraction into a single radicand, so it is finished before any root is taken; the root of a difference is not the difference of the roots.***

His answer is half the true value. Ask him to try $\sqrt{25-9}$ *: his method gives 2 and the value is 4.*

11. Which is the value of $\sqrt{8^2 + 36}$?

- ○ A. 14
- ● **B. 10**
- ○ C. 100
- ○ D. 50

B

12. Select ALL of the expressions whose value is 4.

- ☐ A. $\sqrt{2^3 + 17}$
- ☒ **B.** $\sqrt[3]{64}$
- ☐ C. $\sqrt[3]{-64}$
- ☒ **D.** $\sqrt{16}$
- ☒ **E.** $\sqrt{5^2 - 9}$
- ☒ **F.** $\sqrt[3]{3^3 + 37}$

B, D, E and F

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

Independent Set 4.06 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. $30 - 4 \cdot 5 + 2$

12

2. $36 \div 6 \cdot 2$

12

One level, left to right.

3. $3 + 5(6 - 2)^2$

83

$3 + 5(16)$.

4. $\frac{5^2 - 1}{2^2 + 2}$

4

24 over 6.

5. $-4^2 + (-4)^2$

0

6. $\sqrt{100 - 19} + 2^{-2}$

$\frac{37}{4}$

Under the bar: 81, whose root is 9. Then $9 + \frac{1}{4}$.

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

Question Bank 4.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $20 - 3 \cdot 4 + 2$

10

Multiply first, then left to right: 20 − 12 + 2.

2. $24 \div 4 \cdot 3$

18

One level, left to right: 6 · 3.

3. $5 + 2(7 - 3)^2$

37

Parentheses, then the power, then the product: 5 + 2(16).

4. $\frac{4^2 + 8}{2^3 - 5}$

8

24 over 3.

5. $-3^2 + (-3)^2$

0

The first exponent is attached to the 3 alone; the second is attached to −3.

6. $\sqrt{100 - 36} + 2^{-2}$

$\frac{33}{4}$

Under the bar first: 64, whose root is 8. Then $8 + \frac{1}{4}$.

For questions 7 and 8, evaluate. Each expression has more than one grouping symbol.

7. Evaluate each expression.

a. $4\left(8.63-(3.2)^2\right)+\sqrt[3]{125}$

−1.44

$4(-1.61)+5$.

b. $\frac{2^3-5}{6^2}-\sqrt{\frac{4}{9}-\frac{1}{3}}$

$-\frac{1}{4}$

$\frac{1}{12}-\frac{1}{3}$.

c. $12\left(\left(\frac{1}{2}\right)^3+3^{-2}\right)$

$\frac{17}{6}$

$12\cdot\frac{17}{72}$ *. Written as* $\frac{51}{18}$ *it is the same number.*

8. Evaluate each expression.

a. $\left[3+(2-5)^2\right]\cdot 2^3$

96

Innermost first: $(-3)^2=9$ *, so the bracket is 12.*

b. $\frac{(-4)^3}{2^4}-\sqrt{81}$

−13

$-4-9$.

c. $\sqrt[3]{-27}\cdot\left(\frac{1}{3}\right)^{-2}$

−27

$-3\cdot 9$.

Working backwards.

9. Find the value of n that makes each statement true.

a. $n+3\cdot 4=20$

n = 8

b. $(n+1)^2=49$, where n is positive

n = 6

c. $\frac{n-4}{2^2}=5$

n = 24

$n-4=20$.

Reading someone else's work.

10. Deshawn says that $12 \div 3 \cdot 2 = 2$, because you always multiply before you divide. Is he right? Explain, and give the correct value.

He is not right. Multiplication and division are one level and are taken left to right, so $12 \div 3 = 4$ ***and then*** $4 \cdot 2 = 8$ **.**

There is no rule that puts multiplication before division; the two share a level, and so do addition and subtraction.

11. Which is the value of $18 - 2(3)^2$?

- ○ A. 12
- ○ B. 144
- ○ C. −18
- ● **D. 0**

D

12. Select ALL of the expressions whose value is 4.

- ☐ A. -2^2
- ☒ **B.** $\left(\frac{1}{2}\right)^{-2}$
- ☒ **C.** $2 + 3 - 1$
- ☒ **D.** $\sqrt{16}$
- ☒ **E.** $\frac{6^2 - 4}{2^3}$
- ☐ F. $12 \div 3 \cdot 2$

B, C, D and E

GRADE 7 ACCELERATED · UNIT 4 · LESSON 6

Order of Operations with Radicals

Question Bank - Additional 4.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $30-4\cdot 6+5$

11

30 − 24 + 5.

2. $48\div 8\cdot 2$

12

One level, left to right: 6 · 2.

3. $7+3(9-5)^2$

55

7 + 3(16).

4. $\frac{5^2+11}{3^2-3}$

6

36 over 6.

5. $-5^2+(-5)^2$

0

The exponent attaches to its own base; the parentheses decide what that base is.

6. $\sqrt{169-25}+3^{-1}$

$\frac{37}{3}$

Under the bar: 144, whose root is 12.

For questions 7 and 8, evaluate. Each expression has more than one grouping symbol.

7. Evaluate each expression.

a. $5\left(7.42-(2.6)^2\right)+\sqrt[3]{64}$

7.3

$5(0.66)+4$.

b. $\frac{3^3-3}{4^2}-\sqrt{\frac{5}{9}-\frac{4}{9}}$

$\frac{7}{6}$

$\frac{3}{2}-\frac{1}{3}$.

c. $18\left(\left(\frac{1}{3}\right)^2+2^{-1}\right)$

11

$18\cdot\frac{11}{18}$.

8. Evaluate each expression.

a. $[5+(3-7)^2]\cdot 2^2$

84

Innermost first: $(-4)^2=16$ *, so the bracket is 21.*

b. $\frac{(-5)^3}{5^2}-\sqrt{121}$

−16

$-5-11$.

c. $\sqrt[3]{-8}\cdot\left(\frac{1}{2}\right)^{-3}$

−16

$-2\cdot 8$.

Working backwards.

9. Find the value of n that makes each statement true.

a. $n+5\cdot 3=27$

n = 12

b. $(n-2)^2=64$, where n is greater than 2

n = 10

c. $\frac{n+6}{3^2}=4$

n = 30

$n+6=36$.

Reading someone else's work.

10. Liesel says that $10-4+3=3$, because addition comes before subtraction. Is she right? Explain, and give the correct value.

She is not right. Addition and subtraction are one level and are taken left to right, so $10-4=6$ ***and then*** $6+3=9$**.**

The same answer as Deshawn's, one level down: no rule puts either of a pair ahead of the other.

11. Which is the value of $25-3(2)^3$?

- ○ A. 7
- ○ B. −191
- ● **C. 1**
- ○ D. 6,859

C

12. Select ALL of the expressions whose value is 9.

- ☒ **A.** $\left(\frac{1}{3}\right)^{-2}$
- ☐ B. $36\div 2\cdot 2$
- ☒ **C.** $4+6-1$
- ☒ **D.** $\sqrt{81}$
- ☐ E. -3^2
- ☒ **F.** $\frac{7^2-4}{5}$

A, C, D and F

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 7–8

Name
Date
Period

Real-World Order of Operations

Independent Set 4.07 **ANSWER KEY**

Six questions, on your own, in silence. Show the step that does the work.

1. $6x^2$ when $x = -2$

24

$6(-2)^2 = 6(4)$.

2. $\frac{1}{2}mv^2$ when $m = 90$ and $v = 4$

720

$45(16)$.

3. $\frac{a^2 - b}{c}$ when $a = 9$, $b = 6$, $c = 5$

15

Finish the numerator first: $81 - 6 = 75$.

4. A square room of side 10 feet is carpeted at 2.65 dollars a square foot, with a flat 55 dollars for fitting. Find the total cost.

320 dollars

$100(2.65) + 55$.

5. $2(x+y)^2 - 3$ when $x = 4$ and $y = 1$

47

$2(25) - 3$.

6. A gym charges 25 dollars to join plus 8 dollars for each of 6 classes. Find the total cost.

73 dollars

$25 + 8(6)$ *— only the class price is repeated.*

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 7–8

Real-World Order of Operations

Question Bank 4.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $4x^2$ when $x = -3$

36

$4(-3)^2 = 4(9)$.

2. $\frac{1}{2}mv^2$ when $m = 50$ and $v = 6$

900

$25(36)$.

3. $\frac{a+b^2}{c}$ when $a = 6$, $b = 5$, $c = 2$

$\frac{31}{2}$

The bar groups the whole numerator: $6 + 25 = 31$.

4. $2(x-y)^2$ when $x = 3$ and $y = 7$

32

$2(-4)^2 = 2(16)$.

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 11 feet is floored at 3.20 dollars a square foot, with a flat 75 dollars for delivery. Find the total cost.

$11^2(3.20) + 75 = 462.20$ ***dollars***

The rate is per square foot, so it multiplies the area; the delivery is flat, so it does not.

6. A rectangular deck 14 feet by 9 feet is stained at 1.15 dollars a square foot, with a flat 48 dollars for equipment hire. Find the total cost.

$14(9)(1.15) + 48 = 192.90$ ***dollars***

The area is 126 square feet.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 150$, $h = 65$

25.0

$\frac{105{,}450}{4{,}225} = 24.958\ldots$

b. $w = 180$, $h = 70$

25.8

$\frac{126{,}540}{4{,}900} = 25.824\ldots$

c. $w = 200$, $h = 72$

27.1

$\frac{140{,}600}{5{,}184} = 27.122\ldots$

8. Kinetic energy is $K = \frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m = 1{,}200$, $v = 15$

135,000 joules

b. $m = 850$, $v = 12$

61,200 joules

c. $m = 70$, $v = 8$

2,240 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2 = 900$ and $v = 6$. Find m.

m = 50

$\frac{1}{2}m(36) = 900$, *so* $18m = 900$.

b. $x^2 + 5 = 41$, where x is positive. Find x.

x = 6

c. A square lawn of side n feet is turfed at 2.50 dollars a square foot with a flat 40 dollars delivery, and the total is 2,290 dollars. Find n.

n = 30

$n^2(2.50) = 2{,}250$, *so* $n^2 = 900$.

Reading the situation.

10. Two problems use the numbers 20, 3 and 5. **A:** a gym charges 20 dollars to join plus 3 dollars for each of 5 classes. **B:** a gym charges 20 dollars for each of 3 months, plus 5 dollars to join. Find both totals, and explain why the words 'plus' and 'each' do not decide the operation.

A is $20+3(5)=35$ ***dollars. B is*** $20(3)+5=65$ ***dollars. Both problems contain the same numbers and the same two words, so the words cannot be what decides; what decides is which amount is repeated.***

This is the state guide's named misconception, tested directly: circling the numbers and boxing the key words produces identical markings on two problems with different answers.

11. A square yard of side 15 feet is turfed at 4.80 dollars a square foot, with a flat 120 dollars for delivery. Which expression gives the total cost?

○ A. $15(4.80)+120$
● **B.** $15^2(4.80)+120$
○ C. $15^2(4.80+120)$
○ D. $15^2+4.80+120$

B

12. Select ALL of the expressions whose value is 45 when $x=3$.

☐ A. $(5x)^2$
☒ **B.** $\frac{135}{x}$
☒ **C.** $5x^2$
☒ **D.** $3x^2+18$
☐ E. $5x \cdot 2$
☒ **F.** x^2+36

B, C, D and F

GRADE 7 ACCELERATED · UNIT 4 · LESSONS 7–8

Real-World Order of Operations

Question Bank - Additional 4.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $5x^2$ when $x=-4$

80

$5(-4)^2=5(16)$.

2. $\frac{1}{2}mv^2$ when $m=30$ and $v=10$

1,500

$15(100)$.

3. $\frac{a^2+b}{c}$ when $a=8$, $b=4$, $c=17$

4

The bar groups the whole numerator: $64+4=68$.

4. $3(x+y)^2$ when $x=-5$ and $y=2$

27

$3(-3)^2=3(9)$.

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 13 feet is floored at 2.75 dollars a square foot, with a flat 90 dollars for delivery. Find the total cost.

$13^2(2.75)+90=554.75$ ***dollars***

6. A rectangular patio 16 feet by 7 feet is sealed at 1.40 dollars a square foot, with a flat 35 dollars for equipment hire. Find the total cost.

$16(7)(1.40)+35=191.80$ ***dollars***

The area is 112 square feet.

Two formulas.

7. Body mass index is $B=\frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w=120$, $h=62$

21.9

$\frac{84{,}360}{3{,}844}=21.945\ldots$

b. $w=165$, $h=68$

25.1

$\frac{115{,}995}{4{,}624}=25.085\ldots$

c. $w=210$, $h=74$

27.0

$\frac{147{,}630}{5{,}476}=26.959\ldots$

8. Kinetic energy is $K=\frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m=1{,}500$, $v=20$

300,000 joules

b. $m=640$, $v=9$

25,920 joules

c. $m=55$, $v=4$

440 joules

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2=1{,}600$ and $v=8$. Find m.

m = 50

$32m=1{,}600$.

b. $x^2-7=74$, where x is positive. Find x.

x = 9

c. A square lawn of side n feet is turfed at 3.00 dollars a square foot with a flat 55 dollars delivery, and the total is 1,255 dollars. Find n.

n = 20

$n^2(3.00)=1{,}200$, so $n^2=400$.

Reading the situation.

10. Two problems use the numbers 8, 12 and 4. **A:** a printer costs 8 dollars to hire plus 12 dollars for each of 4 days. **B:** a printer costs 8 dollars for each of 12 reams, plus 4 dollars for delivery. Find both totals, and explain why circling the numbers would not have told you what to do.

A is $8+12(4)=56$ ***dollars. B is*** $8(12)+4=100$ ***dollars. The two problems carry the same three numbers, so the numbers cannot be what decides; what decides is which amount is repeated and how often.***

Ask which number is a rate in each problem — that question has a different answer in A and in B, and no marking on the page can show it.

11. A square court of side 20 feet is resurfaced at 3.60 dollars a square foot, with a flat 200 dollars for the equipment. Which expression gives the total cost?

○ A. $20^2(3.60+200)$
○ B. $20(3.60)+200$
● **C.** $20^2(3.60)+200$
○ D. $20^2+3.60+200$

C

12. Select ALL of the expressions whose value is 32 when $x=4$.

☒ **A.** x^2+16
☒ **B.** $\frac{128}{x}$
☐ C. $2x\cdot 2$
☒ **D.** $x^2\cdot 2$
☐ E. $(2x)^2$
☒ **F.** $2x^2$

A, B, D and F